Proposal Title: European Sustainable Bio-based nanoMAterials Community	
Proposal Acronym: BIOMAC	

HISTORY OF CHANGES

Date	Revision	Changes	
2/10/2020	1	**Part A**	**Part B**
		Changed the title of D2.4 from "Report on upgrading of characterisation and validation testing" to "Report on process optimization"	Minor spelling and grammar mistakes correction
		Section 3.4, table 3.4a Summary of staff effort (person months). Small change of the sum of total PMs to correct addition mistakes. The total PMs were changed from 2094 to 2100.	Section 4.2: Update of 3rd parties' information
12/10/2020	2	Replacement of the SME IDENER (PIC 972790400) with IDENER R&D (PIC 901322643). Because of this change, the total budget was reduced.	Section 4.1.18: Change of IDENER from SME to R&D
		D9.5 changed from DEC to R.	Table 3.1: updated to include more detailed cost breakdowns and CFS costs.
		Deliverable added in WP3: D3.7 Report on the characterization and evaluation process of the demonstrators	Section 3.4: The total budget of the project was changed to 16.596.702,50€ and the funding to 14.807.315 € because of the replacement of IDENER SME from IDENER R&D.
19/10/2020	3	Change of D3.7 from PU to C.	Section 4.2.1: Change of in-kind contributions for ETA from Y to N.
16/2/2021	4	Replacement of BEECO with MISC and transfer of the budget and effort.	Replacement of BEECO with MISC
		Correction of D1.1 responsible beneficiary to IDE	Table 3.4: Change of the equipment description of DTI, typo-error correction of the equipment description of AUTH.
		WT6 Summary of project effort in person-months: Increase of PMs of POLIMI to 150, update and redistribution of POLIMI PMs to the WPs.	Replacement of the image in Section 1.3.1.2, TeC5.
		Tasks 2.3 and 2.4: moving of text from T2.3 to T2.4.	
		Abbreviation of beneficiary BIO BASE EUROPE PILOT PLANT VZW was changed from BBEU to BBEPP to match Part B.	Correction of grammatical and typographical errors.
		Deletion of unit costs of ABIS.	
		Correction of grammatical and typographical errors.	
12/5/2021	5	Addition of new beneficiary (IBB) to represent the SEP. WT6 Summary of project effort in person-months: Revision of PMs of EUBP and IBB. Budget redistribution between EUBP and IBB	Update of Table 3.4.
		Delay of D2.1 from M8 to M10.	Addition of new beneficiary (IBB) to represent the SEP. Revision of the role of EUBP.
		Delay of MS3 from M5 to M10.	Update of Section 4 to include new beneficiary (IBB).
		Correction of grammatical and typographical errors.	Correction of grammatical and typographical errors.
28/6/2021	6	Transfer of person-months dedicated to the Validation Technical Manager position from LIST to AUTH.	Replacement of Validation Technical Manager Youssef Habibi (LIST) by Konstantinos Triantafyllidis (AUTH) and Ecosystem Technical Manager Roberto Lando (STAM) by Giorgia Eranio (STAM).
		Delay of MS4 to M16.	Update of Figure 3.2: Gantt chart

TABLE OF CONTENTS

1. EXCELLENCE 2
1.0 INTRODUCTION AND RATIONAL TO BIOMAC 2
1.1 OBJECTIVES 3
1.2 RELATION TO THE WORK PROGRAMME 3
1.3 CONCEPT AND METHODOLOGY 5
1.4 AMBITION 19
2. IMPACT 23
2.0 BIOMAC IMPACT AT THE LEVEL OF OPERATION 23
2.1 EXPECTED IMPACTS 24
2.2 MEASURES TO MAXIMISE IMPACT 28
3. IMPLEMENTATION 40
3.1 WORK PLAN — WORK PACKAGES, DELIVERABLES 40
3.2 MANAGEMENT STRUCTURE, MILESTONES AND PROCEDURES 42
3.3 CONSORTIUM AS A WHOLE 47
3.4 RESOURCES TO BE COMMITTED 49
4. MEMBERS OF THE CONSORTIUM (BENEFICIARIES) 52
4.1 PARTICIPANTS 54
4.2 THIRD PARTIES INVOLVED IN THE PROJECT (INCLUDING USE OF THIRD PARTY RESOURCES) 162
5. ETHICS AND SECURITY 164

1. Excellence

1.0 Introduction and rational to BIOMAC

Renewable resources are important driving alternatives of economic growth in many industrial sectors, holding promises for sustainable development, unlike conventional petroleum-based ones. Bio-based products derived from biomass can pass through their entire life cycle alleviating obstacles associated with solid waste disposal, supporting at the same time the integration with a circular economy-based material loop. Yet, technical performance, economic feasibility and environmental sustainability should be further enhanced to meet new market requirements. [1] Nano structured bio-based materials (NBM) especially are expected to become highly innovative materials with a great impact on European markets. ***Nanofibrillated cellulose (NFC), cellulose nanocrystals (CNC), bacterial nanocellulose* (*BNC*) and *nanolignin* (*NL*),** which can be utilized as functional nano-additives of bio-based polymers, have progressively attracted considerable interest due to their abundance and potential application in multiple industrial sectors. The markets addressed embrace a wide spectrum of industrial fields such as packaging, automotive, printed electronics (PE), construction, coatings, and products for agriculture, covering strategic sectors of growing industry. A characteristic example is that of nanocellulose (NC), where the market is projected to register a Compound Annual Growth Rate (CAGR) of more than 18% during the period of 2019-2024. [2] Further to this, the global lignin market size was valued at 793.7 million Euros in 2018 and is projected to expand at a CAGR of 1.9% from 2019 to 2025. [3] Constant technological development integrating lignin with nanotechnology is expected to drive the industry further, with the resulting lignin nanomaterials to be utilized as a foundation to produce enhanced high added value end products. To sustain a viable plan, meeting the market's demands, it is necessary to **upscale the supply chain of biopolymers production.** In order to further enhance the quality and performance of these polymeric matrices, the incorporation of nanoparticles (NPs) is essential to achieve advanced properties thanks to their functionalities. It is worth mentioning that the major challenge in the area lays at the fact that new product concepts are restricted to enter the market because end user applications **remain at small lab scale tests,** having limited exploitation capacity for industrial deployment. To launch innovation within this growing sector, BIOMAC proposes the establishment of a self-sustainable ecosystem (Open Innovation Test Bed (OITB)) for upscaling of processes incorporating major developments in this area based on two pillars: to generate knowledge and to capture business value. Offering an open ecosystem, BIOMAC focuses on the creation and demonstration of novel manufacturing supply and value chains, where technologies that have been developed up to TRL 4-5 will be upscaled and validated to TRL 7, accelerating the process towards the market entrance. At the same time, it targets to tackle obstacles that hinder the evolution of concept development in the field of nano-enabled bio-based materials and products, providing a unique opportunity of testing ideas in a well-structured open environment. The BIOMAC ecosystem will act as a cluster of parallel actions that will be assembled in a structured manner, deploying the classification of actions in four major groups of hubs. The core of the ecosystem consists of four Hubs (H). (1H) **Pilot Plant Supreme Hub (PPSH)** that includes three Pilot Lines Clusters (PLCs) covering the whole value chain, from biomass fractionation *(Biomass Fractionation & Pre-treatment Cluster* - ***BFPC)*** to intermediate chemicals and polymers *(Intermediate Material and Nanocomposite Cluster – **IMNC)*** and final nano-enabled biopolymers *(Final Products & Formulation Cluster* - ***FPFC)***. Subsequently three more hubs surround the supreme hub, with the aim of assessing the feasibility and sustainability of the developed concepts: (2H) **Validation Services Hub (VSH)** where the services justify the feedstock, the technologies and the products *(Quality Control Hub and Process Validation Hub)*, the (3H) **Market Uptake Hub (MUH)** that targets to support business perspective and legalization issues accompanied by the maintenance of the test cases, and finally the (4H) **Value Chain assessment Hub (VCH)** where the developed value chains will be assessed against economic and environmental criteria within the circular economy frameworks. The goal of BIOMAC is to provide open access PLs and services that will lower the cost of upscaling and accelerate the time-to-market for nano-enabled biomaterials, enhancing industrial sustainability and reducing waste following a circular model. To reach these targets the ecosystem will function as a **One-Stop-Shop (OSS)** accessible at fair conditions and costs through the collaboration of the multidisciplinary consortium. This will be achieved through a Single-Entry Point (SEP) which will be represented by the well-known and recognized **Industrielle Biotechnologie Bayern Netzwerk GmbH (IBB Netzwerk GmbH, IBB),** a network and service company in the field of Industrial Biotechnology and sustainable economy**,** handling all the business opportunities occurring at the BIOMAC ecosystem and leading the communication towards the clients. Additionally, IBB has access to a significant network of decision makers in the world of bioplastics and a contact database comprising hundreds of contacts which will

Figure 1.1: *BIOMAC Ecosystem*

[1] Can Nanotechnology Improve the Sustainability of Bio-based Products? Journal of Industrial Ecology, Volume 12, Issue 3

[2] https://www.researchandmarkets.com/reports/4745453/nano-cellulose-market-growth-trends-and

[3] https://www.grandviewresearch.com/industry-analysis/lignin-market

facilitate the awareness of BIOMAC. An internal mechanism to set workflows will be handled by the core partners and a single offer will be defined to serve sustainably the clients' requests.

1.1 Objectives

A. Objectives related to the set-up and upgrade of the pilot line Hubs

OA1: To set up an OITB consisting of three Clusters of pilot lines (PLs) offering services in the field of nano-enabled bio-based materials. (PPSH – BFPC, IMNC & FPFC)

OA2: To setup and build a transversal supporting service mechanism that will ensure a fast, reliable, and cost-effective upscaling of bio-based material concepts. (VSH, MUH, VCH)

OA3: To upgrade and appropriately modify the existing PLs through integration of monitoring and process control systems to cope with the biomass variability and to offer continuous and stable conversion of biomass towards the targeted bio-based monomers, biopolymers and nano-additives. (17 PL owners & VSH)

OA4: To develop a holistic model of the proposed processes by demonstrating the link between characterization data and modelling/simulation approaches. (VSH, MUH)

OA5: To validate the operation of the OITB through 5 selected Test Cases (TeCs) (DIAD, NVMT, EVER, ACC & OHM – see Table 1.1.), and 5 more that will be introduced to the OITB through and open call. All TeCs will provide feedback and recommendations on the services provided by the OITB.

B. Objectives related to the operation & sustainability of the ecosystem

OB1 To establish a Single-Entry Point (SEP) platform consistently with a One-Stop-Shop framework. Launch and operate an easily accessible platform in order to attract business cases entering the ecosystem as clients, during and post project. (MUH, VCH)

OB2: To deliver key exploitable results (KER) and organize the Intellectual Property Rights (IPR) protection strategy (e.g., patents, industrial secrets, trademarks). Develop an IPR protection plan and a Data Management Plan (DMP) for the ecosystem. (VCH)

OB3: To tap into niche markets, based on R&D expenditures and CAGR. Identify and penetrate industrial markets, by customer outreach (e.g., advertising, communication, participation in forums and fairs etc.). (All Partners)

OB4: To continuously update the OITB business plan during the project. Overview of EU Research Policy, technology transfer/offer to stakeholders in both directions (technology push, and/or demand pull, stakeholder access). (VSH, MUH, VCH)

OB5: To ensure the post-project sustainability of the ecosystem and create a sustainability plan attracting post project turnover of five times the EU investment within the first five years after the completion. (28 Core Partners)

C. Objectives for the assessment of environmental, economic sustainability & circularity of the ecosystem

OC1: To deliver production processes and products based on highest efficiency and lowest possible environmental impact and energy requirements throughout their life cycle by means of Social-LCA/LCA/LCC (Life Cycle Assessment/Life Cycle Cost). (VSH, MUH, VCH)

OC2: To develop a Decision Support Tool (DST) for the assessment of the value chain developed within the frames of the ecosystem prioritizing options and optimizing processes simultaneously against different criteria. (VCH)

OC3: To validate the circularity of value chains of the ecosystem providing alternative solutions to shift towards circular business models, including End of Life (EoL). (VCH)

OC4: To demonstrate the incorporation and update of nanosafety, regulatory, standardization, and training in the ecosystem, following EU (CEN, REACH) and international regulations and standards (ISO, ASTM). (MUH, VCH)

OC5: To ensure long–term compliance with monitoring and nanosafety procedures. Establish communication networks with the nanosafety cluster (NSC), the European Materials Modelling Council (EMMC) and the European Materials Characterization Council (EMCC) (by developing personal contacts, on-line tools, and joint events). (All)

OC6: To define technical and managerial guidelines that will allow the functioning of the nano enabled bio-based materials for the EoL management chains within a circular network through a closed loop system. (MUH, VCH)

1.2 Relation to the work programme

Specific challenge: Nano-enabled bio-based materials can contribute to a stronger circular economy. nano ….

The exploitation of biomass towards a spectrum of bio-based products is not new. However, the concept is facing many challenges such as heterogeneity of the biomass feedstock, design of sustainable conversion (bio)chemical processes, as well as compatibility of conventional inorganic nanomaterials with bio-based polymeric matrices, thus offering many possibilities for further development and upscaling. BIOMAC addresses the production of biopolymers as well as nanomaterials, across the entire Value Chain, which will enable optimized design of a wide range of nanostructured materials with the goal to improve the overall circularity of the end products. Nanomaterials such as NFC, CNC, BNC, and NL will be obtained from the earliest stages of the value chain of biomass processing and fractionation to cellulose, hemicelluloses and lignin, followed by production of sugar-and aromatic- derived building blocks for the synthesis of biopolymers and their nanocomposites reaching the formulation of the final marketable products. BIOMAC starts with the pre-treatment and fractioning of biomass in a semi-continuous mode that can use multiple raw materials simulating an industrial environment. At first it will be concentrated on lignocellulosic crop (*Miscanthus*) biomass (MISC), securing the availability of feedstock and continue processing paths without seasoning dependence. However, the processes are agnostic, so both agricultural and forest residues are candidates to be used,

so the European Biomass Industry Association (EUBIA) will contribute to open channels with existing value chains. Nanocellulose (NC), Nanofibrillated cellulose (NFC), Nanolignin (NL) and Bacterial nanocellulose (BNC) will be used in the biopolymeric matrices in order to impart new functionalities with the ultimate goal of achieving bio-based materials with enhanced performance in terms of functional (e.g., thermal stability, flame retardancy, antioxidant properties, smart sensing PE, antibacterial/microbial) and mechanical behaviour in the medium to long-term. Sustainable precursor monomers such as lactic acid (LA), succinic acid (SA) as well as different saccharides, polyols (such as oligosaccharides from hemicellulose, sorbitol, xylitol, ethylene glycol (EG), 1,3-Propanediol (PDO), 1,4-butanediol (1,4-BDO), depending on the targeted polymers) will be produced in order to obtain biopolymer matrices such as PLA and its copolymers, succinate-based polyesters and resins, such as isocyanate free thermoplastic polyurethanes (TPU) and polyurethanes (PU). Closing, it is worth mentioning that around 9% percent of plastics entering the waste stream are recovered for recycling, though recovery rate is not necessarily indicative of the final recycling rate. BIOMAC will take into significant consideration the environmental impacts by promoting the use of green technologies to meet the increasing demand for these materials without compromise to **Product Quality, Durability & Performance or Economics**. Also, **end-of-life** (EoL) of the final products will be consistently addressed, for example products for agricultural applications biodegradable in soil are themselves a perfect case-study of the circular economy, as all the products originating from biomass will be incorporated into biomaterials designed to fully biodegrade in soil providing fertile carbon and also closing the carbon loop back into the soil.

Scope: Establish Open Innovation Test Beds (OITB) by upgrading existing or developing new materials ….

BIOMAC aims to create a self-sustainable open access ecosystem consisting of **17 open access** PLs across the entire bio-based value chain comprising of the production and incorporation of bio-based nanoparticles (NC, NL & BNC) in the final products. Within **the Pilot Plant Supreme Hub (PPSH) 3 Clusters of Pilot Lines** (BFPC, IMNC and FPFC) will be subjected to modifications such as process control, larger scale application, batch to continuous mode, nanomaterial compatibility mixing and integration, process & material simulation together with monitoring systems ensuring more efficient process control and optimization. Supported by characterisation facilities, modelling tools, regulatory and standardization consultancy and health and safety (H&S), all cooperating within the framework of innovative business models, that will be implemented internally and provided as services. The assessment services include the consideration of s-LCA / LCA and LCC along with the development of DST towards the screening of all the possible options using simultaneously environmental, economic and energy criteria. To provide sustainable and holistic production paths, BIOMAC will offer dedicated services examining the circularity and EoL advancing towards circular approaches and business models. **Regulatory, safety and environmental barriers** will be confronted through active involvement in standardization and regulatory committees at a national and European level.

The focus is on new nano-enabled bio-based materials' functions, features nano-enabled bio-based …

The network of PLs in BIOMAC aims to develop tailor-made solutions for producing and integrating nanomaterials across the bio-based value chain, that will accelerate smart functionalities and enhanced properties of biopolymers, as aforementioned. Depending on the final application, a number of novel biopolymers, inks, coatings and resin materials will be developed and validated within the PL Hubs. Each material will provide several tailored properties suitable for the supply chain, targeting end applications and associated Test Cases (TeCs) see 1.3.1.1. Furthermore, **EUBP is licensor of the Seedling® logo**, an EN 13432-based conformity mark of industrial compostability for biodegradable plastic packaging. EUBP has created and owns the Seedling® mark of conformity for compostability of plastic packaging. EUBP continues to be involved in the supervision of the EN 13432 standard upon which European certificates of biodegradability of plastic packaging are based.

Proposals should assess regulatory, safety, economic and technical barriers and should develop and validate ….

BIOMAC will confront the main barriers for biomass processing and nano reinforcement, offering therefore to fill the gap between development and market penetration readiness. Two main barriers have been identified: **Quality control** (QC) and **Process Validation** (PV) **barriers** become a crucial factor for the upgrade of the PL Hubs. **Technical Barriers** consist of limited access to large scale pilot lines for biomass pre-treatment and processing, as well as lack of capacity to evaluate products in large scale. These will be surpassed by the overall structure of the ecosystem that addresses a multidisciplinary field. The **Exploitation barriers,** such as the high risk for investors to adopt non-validated and non-standardized technologies etc. and low profit margins leading to limited resources to invest in R&D, will be overcome via continuous innovation management and securing the Freedom to Operate (FTO), meeting real market needs. EUBP is a member of several CEN and ISO committees dealing with plastics and packaging, providing authoritative input for all relevant standards on the European level which will help to the full-scale exploitation of the new industrial products.

Open access at fair conditions and cost as well as outreach and dissemination across Europe, based on …..

BIOMAC will provide open access to clients through a SEP, particularly SMEs, giving them the opportunity to design, upscale and test biomaterials at a fair cost, without investing huge budgets for the upgrade of PLCs beforehand. Industrial requirements will be clearly defined and stated, technology road mapping will help dealing with the increasingly competitive markets, followed by business planning and market mapping, while competitive strategies and Intellectual Property (IP) tracking will also be established. To facilitate these actions, a web platform, acting as a virtual SEP will developed. The platform will be dedicated to clients to support their involvement in the OITB with

the submission of their test cases (RDC). BIOMAC will liaise with SMEs, MidCaps, Stakeholders, and investors community to commercialize products deriving from the multiple supply chains through its dissemination channels (website, fairs/workshops, communication material & training). The Innovation and Exploitation will facilitate SMEs access to funding opportunities and access to additional funding (Venture capitals and/or Angel investors).

Materials should be demonstrated in relevant industrial environments.

Test and Demonstration activities in an industrial environment will validate the proper Value Chain and Product development operation. In particular, 5 TeCs covering different industrial fields relevant in the European market, such as those of packaging, automotive, printed electronics, construction, and agriculture will be addressed and demonstrated. A detailed presentation of the end products is included in Table 1.1. These TeCs will be used as lighthouse case studies to attract additional clients. Moreover, through an open call for access, five more TeCs will enter the OITB at **Month 33**. All 10 TeCs will lead to prototypes that will be demonstrated and validated in TRL7 (operational environment) by the End Users in collaboration with the OTB members.

… to facilitate cooperation across Europe, with other projects … in the course of the project.

The BIOMAC OITB consists of 35 partners from 12 countries across Europe represented by technological organisations with important contribution of industrial players (12 SMEs, 2 Associations and 3 LE). Further to this, there are similar initiatives (previously funded OITB & BBI) and established European/National clusters that partners of the BIOMAC consortium are already involved in (e.g., EPPN, Nanosafety, TireMat- IPBG, Luxembourg, Flagship graphene, AFTERLIFE, NanoFutures). BIOMAC will further promote and facilitate cooperation.

Activities should start at TRL 4 and achieve TRL 7 at the end of the … proposals requesting other amounts

BIOMAC has been built around new technological solutions of the entire supply and value chain. TRL4 PLs has been already reached and demonstrated by work conducted previously by the partners, and a total funding of ***14.807.314€*** is required to reach TRL7.

1.3 Concept and methodology

1.3.1 Concept of BIOMAC OITB

BIOMAC intends to build a sustainable OITB consisting of **17 Pilot Plants and supported by various technical and non-technical services for the upscaling of new bio-based materials concepts.** The PLs have been selected based on their versatility and complementarity to deliver nano-enabled biomaterials based on alternative types of biomass, leading to products addressing a large variety of industrial sectors of prominent relevance in the growing market. The ecosystem will secure its proper operation based on an already developed business plan and marketability strategy. A **supporting mechanism in terms of Services Hub** has been set up following a holistic approach that includes process control, validation, Quality Assurance and Quality Control (QA/QC), innovation management, circularity, standardization, certification, nanomaterials handling safety and regulatory aspects that will support the smooth transition of product ideas to the market. (Figure 1.2) After the completion of the project, the OITB will operate via its SEP IBB promoting the collaboration and knowledge transfer to SMEs, levelling the deployment and commercialisation of the results. BIOMAC will be implemented in three phases:

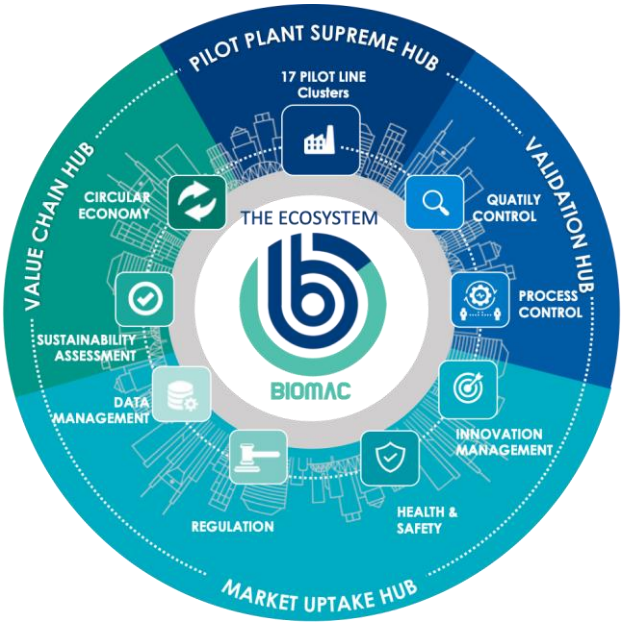


Figure 1.2: *BIOMAC OITB Structure*

Phase 1: Set up of the structural and organizational aspects of BIOMAC. The first phase relates to the organization of the four Hubs, the development of workflows and the definition of internal and external operational procedures. The focus will be given to the proper operation of the Hubs, highlighting the tracking of the value chain, and streamlining the synergies between stakeholders. The ability of innovators to engage with end user community to understand their needs, motivations and issues is critical to develop products that meet the market needs. Based on this fact, the user-driven ecosystem will focus on the interaction with the potential customers (upstream and downstream) as well as on bridging producers with end users. A SEP mechanism will be set up to simplify the access of the end users to the OITB. IBB SEP will have the mandate to sign contracts with clients on behalf of the OITB members. During this phase, all administrative, organizational, and operational aspects will be defined.

Phase 2: Validation of the ecosystem. During the second phase, the ecosystem will be validated in relation to the services that it provides. At first, internal Test Cases (interTeCs) will validate the proper operation (the 5 TeCs already existing in the consortium DIAD, NOVAMONT, EVERSIA, ACC & OHM) and second, an open call for expression of interest for industry and particular SMEs will be launched, inviting 5 more intra Test Cases (intraTeCs) to assess the OITB. Both upstream (materials producers, service providers) and downstream (end users, integrators) industries

will access the ecosystem through the SEP and specific cases will be created based on the needs, specifications, and requirements of each case. During the validation process a variety of case studies will be examined focusing on new value chains.

Phase 3: Ecosystem sustainability. The BIOMAC ecosystem will be set up in order to ensure its future financial sustainability. During the project, appropriate activities will be undertaken towards this direction with special emphasis to the promotion of extroversion. A dissemination and communication plan will be implemented including workshops, marketing activities, and open access of the ecosystem during the Phase 2. Ecosystem business plans will be developed targeting in the acquisition of clients and the creation of additional turnovers and jobs. [4] All this will progressively create the OITB business model and will create opportunities and commercial contracts.

1.3.1.1 Hubs of the BIOMAC OITB

A. Pilot Plant Supreme Hub (PPSH): This hub includes three clusters of PLs for the upscaling of multifunctional nanostructured materials that will be available for open access by the clients of the ecosystem. In particular, the OITB starts from biomass pre-treatment, followed by the production of nanomaterials and intermediate platform chemicals, towards the development of end bio-products offering multiple advantages such as shorter processing time, better selectivity, lower energy and footprint, fewer production steps and reduction of by-products. The ecosystem is based on three clusters of PLs, along the overall value chain:

(i) Biomass Fractionation and Pre-treatment Cluster (BFPC): This group consists of 2 Pilot Lines (PL) (a) Advanced semi continuous Organosolv- Steam Explosion Fractionation (LTU) and (b) hydrothermal treatment (BBEPP). Based on these innovative processes, the Ecosystem will be able to cover the pre-treatment of the most common type of lignocellulosic biomass available in Europe, while intending to additionally validate relevant fractions of biomass. The pre-treatment lines are devoted to fractionating the feedstock allowing the extraction of the most profitable fractions of biomass (i.e., cellulose, hemicellulose and lignin) to be fully valorised across the value chains. These main fractions are serving either as precursors towards the production of intermediate monomers, or for producing nanocellulose/nanolignin/biochar. [5]

(ii) Intermediate Materials and Nanocomposites Cluster (IMNC): produce state-of-the-art platform chemicals such as LA, SA and different polyols/diols (BBEPP, ATB, AUTH) that will be employed as biomass-derived monomers and building blocks to finally produce bio-based polymers and related nanocomposites. Further to these technologies this cluster of PLs involves the production of nano-based materials, such as NFC, CNC, BNC, NL and Biochar [6,7,8] (ITENE, RISE, UEdin, LIST, Cnano). These nanomaterials will be applied along the value chain with a scope of enhancing the properties and functionalities of the bio-based polymers for the targeted end user applications. [9] In addition, the isolated lignin as well as suitable oligosaccharides and small molecular weight lignin fractions generated during the biomass pre-treatment will be incorporated as components with other building blocks to produce fully bio-based and isocyanate free TPU with tailored properties (LIST, FH, AUTH).

(iii) Final Products Production & Formulation Cluster (FPFC): Thermoplastic polymers such as PLA and its copolymers, TPU, polyesters and PU/resins will be processed through reactive extrusion REX (LIST, AIMPLAS), [10] additive manufacturing (AIMEN), inkjet printing (DTI), coating (ITENE, AIMPLAS) and R2R Nanoimprint lithography (NNT). These will cover end user applications for automotive (DIAD), biodegradable in soil agricultural applications (NOVAMONT), food packaging (EVERSIA), construction (ACC) and printed electronics in textile/wearables sector (OHM). The choice of the applications has not only been based on the wide market, but also based on their EoL in the automotive, food, framer of circular economy and to produce eco-friendly products.

PL descriptions, Upgrade & Modification Plan: In order to achieve an efficient exploitation of the PPSHs and accommodate a customized approach, each PL has foreseen necessary modifications and upgrades that will take place where needed (e.g., integration of auxiliary and peripheral systems, upscaling and continuous operation) in order to provide flexibility on the adoption of a variety of bio-based monomers, polymers and nanoparticles. PLs Orange stand for Pre-Treatment, Green for Nanomaterials production and Blue for monomers and nano-enabled bio-based materials. The upgrades scheme is presented:

[4] [EUBP] Bioplastic market data 2019 – Global production capacities of bioplastics 2019 – 2024"

[5] [LTU] P. Christakopoulos et.all. Technoeconomic Assessment of Hybrid Organosolv–Steam Explosion Pretreatment of Woody Biomass 2019, 12, 4206

[6] [LIST] Y. Habibi. Cellulose NC: Chemistry, Self-Assembly, and Applications, Chem. Rev. 2010, 110, 6, 3479-3500

[7] [LIST] Y. Habibi. Key advances in the chemical modification of nanocelluloses, Chem. Soc. Rev., 2014, 5, 1519-1542

[8] [PoliMi] G. Griffini. NL by ultrasonication and their incorporation in waterborne polymer nanocomposites, https://doi.org/10.1002/app.45318

[9] [AUTH] D Bikiaris, Can NP enhance thermal stability of polymers? an overview on thermal decomposition of addition polymers, Therm Acta 523 (1-2), 1-24

[10] [LIST] Y. Habibi, et.all. Polylactide (PLA)-based nanocomposites Progress in Polymer Science, 2013, 38 (10-11), 1504-1542

PL1 (LTU, SE): Semi-continuous organosolv-steam explosion pre-treatment PL

The semi-continuous organosolv-steam explosion pre-treatment PL is capable of fractionating lignocellulosic biomass (e.g., agricultural and forest residues) to its three main polymer components, i.e., cellulose, hemicellulose and lignin. It offers unique advantages by combining the effects of steam explosion with that of hemicellulose and lignin solubilization (organosolv) in ethanol/water mixtures under pressure and temperatures up to 200°C. The semi-continuous PL of LTU provides the additional benefit of solid/liquid separation that can take place during operation at high temperature and pressure, simulating the continuous organosolv operation mode. This process offers unique advantages when compared with the conventional batch process, such as reduction of the specific energy consumption by almost 50%, significant reduction of the lignin content in the extracted cellulose fraction and increased recovery of un-degraded lignin.

Upgrade & Modification Plan: The current semi-continuous set-up will be upgraded to a fully continuous system by adding a continuous biomass feeder with a capacity of 1-3 kg/h of wood chips, which will be continuously introduced to the main reactor while in operation (50-60% w/w ethanol/water as solvent, up to 200°C and up to 30 bar), producing a solid fraction as a press cake, containing mainly cellulose, a liquid fraction (the press juice) containing the solubilized hemicellulose, and lignin.

PL2 (RISE, SE): Hydrolysis of fibre sludge and BNC production.

Fibre sludge (FS), a side stream from the pulp and paper industry will be used as the raw material for BNC production. FS will be enzymatically hydrolysed, and the released sugars will be converted to BNC by a Komagataeibacter. strain isolated in house. Both Hydrolysis and fermentation stages will be carried out in the Pilot park of RISE equipped with fermenters with capacities from 50L to 600L. RISE also manages demonstration scale equipment (m^3 scale) (Biorefinery Demonstration Plant) which is available for enzymatic hydrolysis of FS during the project. Optimisation of operational parameters such as the feeding rate and stirring speed will be performed, targeting increased productivities, yields and titres of BNC. Optimization to define the crucial factors (endogenous or exogenous) for BNC production by will be performed. The factor(s) defined will be used as the target of an evolutionary engineering strategy to be applied in parallel with process optimization, targeting an improved fermentative performance at certain process conditions.

Upgrade & Modification Plan: An upgrade of downstream processing will be performed regarding the continuous mode of operation. The fermenters will be upgraded with all necessary instruments for continuous, fed batch or batch production of BNC.

PL3 (AUTH, GR): Sugar derived polyols and diols by catalytic hydrogenation/hydrogenolysis

The PL will comprise of high-pressure, continuous flow, two packed-bed reactors that can operate in separate or in cascade mode for the initial selective production of polyols/sugar alcohols (sorbitol, xylitol) and the controlled downstream hydrogenolysis to glycols/diols (glycerol, EG, 1,2-PDO), using state-of-the-art commercial or in-house catalysts. This pilot line configuration offers the flexibility to simultaneously operate both processes which require different conditions, e.g., 100-120°C, 40-80 bar H_2 for hydrogenation vs. 180-240°C, 40-80 bar H_2 for hydrogenolysis. Similarly, when sugar derived di-acids (e.g. succinic acid) will be used as feedstock, the production of α,ω-diols (e.g. 1,4-BDO) will be performed via lactone (gamma-Butyrolactone) intermediate formation. The polyols/diols product streams will be used as received in the various polymerization reactions in comparison to pure single compounds towards minimizing the need of separation/purification and increasing the economics of the whole value chain. Still, separation/purification of polyols/diols from the product mixtures will be also developed in PL 6.

Upgrade & Modification Plan: The PL will be upgraded by adding a second high pressure, continuous flow stirred reactor (with a process capacity of 1-20 L/day of 10-20wt.% sugar feed solution, depending on the required Weight Hourly Space velocity (WHSV) with the necessary peripherals, such as catalyst packed-bed holding device, furnace, tubing, mass flow controllers, feeding pumps, high pressure gas/liquid separators and product collectors, alarm shut off valves, control and data acquisition systems. The existing high pressure, continuous flow packed-bed reactor,

operating under trickle-bed mode will be also upgraded with a larger tube reactor (processing capacity 1-20 L/day of 10-20wt.% sugar feed solution).

PL4 (Cnano, GR): Ultrasonication (US) assisted PL reactor for the production of NL

This PL will allow to obtain stable colloidal suspensions of NL particles in the range of 10 to 50 nm by physical treatment avoiding the need of hazardous chemical pre- or post-treatments. In the current PL configuration, the US treatment will enhance the exploitability of lignin as nanostructured functional filler (enhanced thermal stability, antibacterial properties, biocide action) in bio-based polymer formulations and compounds at a 50L batch scale reactor. The flow cell throughput rate will be around 100 L/h – with main variables being viscosity of the dispersion (solids concentration) and desired degree of processing (US frequency and amplitude, power, treatment time, temperature), to selectively control size, size-distribution and chemical characteristics of the resulting NL particles.

Upgrade & Modification Plan: Upscale of US reactor at a frequency higher than 20 kHz, 1500 W power. When used with a flow cell it can run larger sample volumes in recirculation to establish the correlation between process parameters, such as amplitude, temperature and residence time, and the process results (NL particle yield, size and size distribution, chemical functionality). Investments will be done in order to upscale the material throughput and adapt the process conditions of the PL by installing a higher power US probe and by integrating an in-line temperature monitoring system. The process monitoring will be connected to CPI units to remotely finetune the process.

PL5 (BBEPP, BE): Hydrothermal pre-treatment of biomass

Pre-treatment of biomass will be performed in order to obtain lignin, hemicellulose or cellulose rich fractions. These fractions or the related monomer sugar streams after enzymatic hydrolysis will be further processed in PL7 and PL10. Pure sugar streams (i.e., glucose or xylose) in combination with PL6, will serve as substrate to PL3 for catalytic hydrogenation/hydrogenolysis to sugar alcohols (polyols) and diols. Each pre-treatment method works differently in the physico-chemical deconstruction of lignocellulosic structures thus resulting in different oligomeric and monomeric enriched fractions. Two main methods will be applied, both leading to efficient separation of the biomass fractions and increased enzymatic hydrolysis towards monomer sugars. **1st Approach:** Weak acid and hydrothermal pre-treatment which increases the solubility of the hemicellulose, resulting in a hemicellulose rich liquid fraction. The remaining solid fraction contains cellulose and lignin that can be further processed by enzymatic hydrolysis resulting in a glucose rich liquid fraction. **2nd Approach:** Alkaline pre-treatment which increases the solubility of the lignin fraction, resulting in a solid fraction enriched in cellulose and hemicellulose, which can be easily enzymatically hydrolysed to monomeric sugars. For the different biomasses, parameter optimization will be required in order to achieve the desired monomeric characteristics, while limiting the formation of unwanted by-products (inhibitors) such as furfural and hydroxy methyl furfural. All these processes will be performed in batch reactor vessels up to 4 x 8 m³, stem, up to 6 bars. The liquid fraction will be separated from the remaining solid fraction using centrifugal technology, such as decantation, or size exclusion technology (filter press, membrane cross flow filtration, etc.).

Upgrade & Modification Plan: In order to improve biomass fractionation capabilities, implementation of grinding and milling steps before pre-treatment will take place, or an improvement of the pre-treatment capacities will be applied to be able to reach higher temperatures during pre-treatment.

PL6: (BBEPP, BE): Recovery / Separation

The different liquid fractions from different pilot lines, such as PL5 and PL3, will be further purified in PL 6 when needed. This pilot line comprises filtration equipment (microfiltration, ultrafiltration, nanofiltration and reversed osmosis), ion exchange equipment, solvent extraction technology and crystallization to obtain monomeric fractions that then can be further processed in other process lines. Pure monomer sugar streams (i.e., glucose or xylose) will be recovered from the liquids of PL5, while sugar alcohols (i.e., sorbitol) will be separated and recovered from smaller glycols, e.g. EG and PG, from the liquid products of PL3. The pure compounds will be used in the value chain for hydrogenations or polymerizations and will be compared to the performance of using mixed streams, i.e., mixed polyols/diols.

Upgrade & Modification Plan: In order to ensure the purity and recovery of the biomass fractions and materials, optimization tools for simulations and mathematical optimisation will be applied for the optimisation of the processes.

PL7 (ATB, DE): Enzymatic Hydrolysis & Microbial Fermentation for SA and LA

Succinic & lactic acid (SA/LA) from several feedstock streams will be produced by ATB at laboratory and pilot-scale using several fermentation approaches. (1) Enzymatic hydrolysis depolymerizing biopolymers like starch, cellulose to fermentative sugars glucose (C6) and xylose (C5), by means of hydrolytic enzymes; (2) Fermentation metabolizing sugars to LA and SA; (3) Separation and purification of lactic/succinic acid purification to meet the standards of commercial applications; (4) Up-scaling of the entire process chain including downstream processing for the provision of product samples in kg-scale and high quality. Fermentation and biomass composition can be followed conveniently by Near-infrared (NIR) as done by IRIS in prior PERCAL and AgriMax projects.

Upgrade & Modification Plan: A stable continuous process improves the entire productivity, online measurement devices to get quicker results of different steps especially for DSP will be evaluated. At line measurement devices to get quicker results of the different steps from the upstream (hydrolysis) via bioconversion (fermentation) up to downstream processing (separation and purification of LA & SA) will be needed across with process optimization to attain high productivities of the main products (LA, SA) by changing parameters such as carbon source feeding regime to avoid substrate inhibition, the optimum dry matter content and the best nutrient supplementation for several bacteria.

PL8 (UEdin, UK): Pyrolysis and carbonisation of biomass

This PL will process biomass fractions up to 50 kg/h of input material at temperatures up to 850°C, producing porous solid carbons, liquids rich in phenolic compounds, and combustible gases. Solid carbons with high porosity and tuneable surface functionality, and controllable electrical conductivity, suitable for additives for packaging polymers, printed electronics, building materials, animal feed additives and agricultural products (fertilisers and soil amendment) can be produced by this facility. The PL control and monitoring system will be upgraded, an improved feed preparation and feeding system will be added to enable handling of diverse feedstock materials including nanomaterials. In addition, sensors for feedstock composition monitoring and product quality monitoring will be added as well as environmental emissions monitoring system.

Upgrade & Modification Plan: the pilot scale facility will be upgraded to allow higher feedstock flexibility (size, density, shape, etc.) so that difficult to handle materials, e.g. lignin or nanoparticle containing materials, can be processed, advanced process monitoring and control (feedstock composition, material flows, T, gas, etc.), improved product recovery and separation (phenolic compounds, organic acids, etc.), and improved environmental performance (effluent and product gas handling and monitoring) will be implemented. The aim is to enable reliable, efficient, and safe processing into selected final, such as porous solid carbons and phenol-rich liquids up to TRL7.

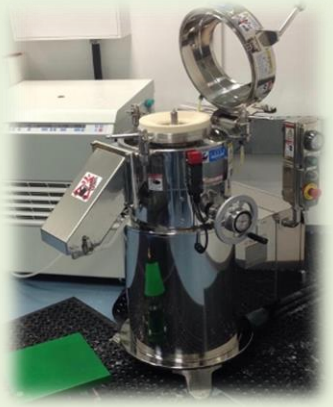
PL9 (LIST, LU): Continuous Reactive Extrusion for TPU

Reactive Extrusion of TPU will be carried out in a twin-screw extruder 18 mm equipped with 6 gravimetric solid feeders which allow a continuous processing. Such process offers many advantages over standard polymerisation processes which are batch and not very flexible in terms of conditions. Indeed, while being easy to control, our continuous reactive extrusion process allows polymers/monomers out-puts from up to 10 kg/h under a variety of specific energy conditions: temperatures up to 450 °C, - screw speed up to 1200 rpm, - large diversity of complementary screw profiles, - length of extrusion from 44 to >100 L/D when combining extruders together in tandem. Besides, when combined to simulation software, it is possible to simulate any scale of the developed twin-screw extruder. At first, it is expected to use a UV/Vis spectrophotometer in the feeder of the lignocellulosic biomass to track phenolic compounds and adjust the process accordingly. A spectrometer equipped with optical probes (expectedly in the NIR range) will be connected to several ports along the extruder to follow in situ the evolution of the degree of polymerization of the non-isocyanate thermoplastic polyurethanes (e.g., tracking the esterification and/or amine conversion).

Upgrade & Modification Plan: Besides the quality of the dispersion of the nanoparticles in the thermoplastic, the online determination of the progress of polymerization is also of interest. Indeed, a FT-NIR spectrometer will be

installed inline to determine in situ the degree of polymerization of the non-isocyanate thermoplastic PU, it is envisaged to set different inline monitoring points along the REX line leveraging the potential of photonic techniques in tailored spectral ranges. A FT-NIR equipped with optical probes will be connected to several ports along the extruder to follow in situ the evolution of the degree of polymerization of the non-isocyanate thermoplastic polyurethanes. Optionally, UV-Vis spectrophotometer will be installed in the feeder to track phenolic compounds and adjust the process accordingly.

PL10 (LIST, LU): Mechanical milling and Production of different grades of nanofibrillated cellulose NFCs



Nanocellulose substrates in the form of nanofibrillated cellulose (NCFs) will be produced through mechanical shearing of pure cellulosic fibres or cellulose fibers with different compositions particularly different content of residual lignin; all of them produced from LTU PL. A SuperMassColloider Grinder from Masuka with treatment capacity of 100 kg/h of cellulose slurries (up to 5% of cellulose content) will be used. This technology allows treatment of different cellulose fibres (short and long) with various compositions at lowest energy consumption and without clogging issues encountered with the other technologies. The treatment can be adjusted to produce NFCs with desired aspect ratio (Length/Diameter). Chemical pre-treatment based on oxidative agents will be used when needed to soften the fibers prior to the mechanical treatment. Post chemical modifications of the resulting NFCs to meet better compatibilsation with the polymeric matrices and/or introduce additional desired functionalities can be also realised. The final NFCs will be incorporated in the polymeric (nano)materials for the targeted applications owing their properties.

***Upgrade & Modification Plan**:* a continuous set-up will be integrated allowing treatment of different amounts enabling thus the screening of different conditions to adapt to different end-use applications and a monitoring system in order to produce constant quality.

PL11 (AIMPLAS, ES): Reactive extrusion (REX) for PLA and PLA copolymer based nanocomposites.

The core of the REX pilot line consists in a twin-screw extruder with 27 mm of screw diameter and L/D ratio of 56, equipped with several side feeders, vacuum venting and liquid injection ports, that will be used as a continuous reactor for polymerization of PLA and its copolymers by Ring Opening Polymerization (ROP) of lactide. The main advantages of the REX process compared to standard batch approaches is a faster polymerization (from hours to a few min), [11] the ability to work in a continuous mode, showing better mixing/dispersion capacity, and making easier to handle high viscosity materials in a free or almost free solvent process. The polymerization of PLA and its copolymers will be carried out in presence of NPs obtained in other PLs, thus allowing the *in-situ* polymerization of PLA nanocomposites to be extruded into pellets. The production capacity will range up to 10 kg/h. REX conditions will be applied in a co-rotating twin screw extruder where the final process parameters (screw and extruder configuration, T profile, throughput, and residence time) will be optimized.

*Upgrade & Modification Plan**:*** Different twin screw extruders are available (L/D ratio 44-56) with several side feeders, vacuum venting, injection ports for liquids/gases, feeders. Software tools to screw simulation will support the process design to achieve productions of 4-8 kg/h. In terms of in line monitoring of REX, a similar approach will be followed as for LIST, adjusting the monitoring points and spectral range to the specificity of the PLA reaction process (e.g., FT-MIR could be needed to obtain a fine tracking of the extent of the reaction.

[11] Jérôme, Robert. et.all. (2000). New developments on the ring opening polymerisation of polylactide. Industrial Crops and Products. 11. 265-275. 10.1016/S0926-6690(99)00053-9.

PL12 (FH-WKI, DE): Resins Pilot Line

In this PL, monomers from PL3, such as succinic acid, glycerol, EG, 1,2-PDO and sugar alcohols will be used as starting materials for bio-based polyester and polyurethane resins. For UV-curing applications, bio-based itaconic acid or acrylic acid will be incorporated into the polymers. The resins will be synthesized by azeotropic polycondensation or polyaddition reactions. Reactors on lab-scale (up to 2 kg) and small pilot scale (up to 4 kg) in a double-walled steel reactor and computer-monitored temperature control for the synthesis of polymer resins will be upscaled. These polymer resins will be used as starting materials for additive manufacturing and printing inks. In addition, the resins can be formulated for coatings and additive manufacturing purposes and applied on different substrates. UV-curing of samples, as well as SLA-machine for additive manufacturing is available.

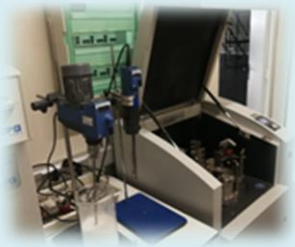
Upgrade & Modification Plan: Investment for a bigger reactor (up to 50 kg) to be able to provide bigger polymer resins volumes necessary for additive manufacturing, coating and printing ink applications envisaged. This will also allow for larger volumes for industrial clients, interested in the services of this during and after the funding period of this project.

PL13 (ITENE, ES): Mechanical treatment to produce NFC and / or CNC

Cellulose will be mechanically disintegrated in a grinder. This grinder will work in continuous mode and viscosity will be inline monitored. It will reduce working times, reagents, and material losses, in order to supply particles in the proper format to the companies. It is expected the production of 1 kg dry MFC, NFC or CNC / day. Surface modification of nanocellulose to enhance its dispersion in polymers, coatings and /or inks will be carried out in vessels (from 20 to 150 litres) equipped with temperature heaters. Where needed, different stirring systems (ultrasounds and turrax) from PL14 will be used. 2kg of dried modified nanocellulose per day will be produced. Finally, for those final applications which would require nanocellulose in dry form, the pilot Spray-drier (GEA) will be used to obtain different particle sizes (50 g/day).

Upgrade & Modification Plan: The grinder will be modified to with an automatic feeding system for continuous flow and inline monitoring, increasing the production of NFC to 2 kg dry NFC/day. On the other hand, acquisition of new disintegration stones will be needed to obtain smaller fibre sizes. The surface modification system will be upgraded through the acquisition of new equipment to improve the dispersion and mixing, increasing the reaction yields. Upscaling of the pilot spray drier to increase the dry nanocellulose production capacity will be done via the modification of its cellulose inlet nozzle.

PL14 (ITENE, ES): Coating Formulation PL



The nanocellulose obtained in the PL13, as well as other bio-based materials from the others PLs will be used in this PL to formulate nano based coatings and inks. This PL will be based in a previous step for mixing and dispersing bio-based material with the other compounds needed in the formulation. equipped with temperature heaters and Different stirring systems equipped with temperature heaters will be used. In case it will be needed, vessels (from 20 to 150 L) from PL13 will be used. For milling, 2 different equipment will be available according to the viscosity of the product to be processed. For those products with a lower viscosity which will be used to the formulation of liquid inks, a Labstar horizontal bead mill will be used. For those products with a higher viscosity, such as coatings or paste inks, a three-roll mill will be used.

Upgrade & Modification Plan: The mixing and dispersion system will be upgraded through the acquisition of new modules and accessories to improve the dispersion and mixing, increasing the reaction yields.

PL15 (AIMEN, ES): Additive manufacturing (AM)

PL15 will cover the value chain from product engineering to the final product manufacturing related to two 3D printing technologies for bio-based materials processing: Fused Filament Fabrication (FFF) for thermoplastic TPU (LIST) polymers and photopolymerization (single-photon polymerization-SPP & multi-photon polymerization-MPP) for bio-based resins (FH-WKI). PL15 consists of a prototype composed by physical infrastructure and a digital process pipeline. Infrastructure for FFF process integrates: 1) Filament extruder for creation of thermoplastic FFF filaments, with customized percentage of nano-additives (including cooling and filament winding); 2) a FFF multi-nozzle and high temperature printing machine enabling a part manufacturing (up to 420 x 590 x 400 mm), and temperatures monitoring; 3) a robotized FFF printing head to extend the building volume and freedom for manufacturing complex geometries. Infrastructure for photopolymerization process implemented in isolated pulsed lasers from picosecond to femtosecond pulse-width, with wavelength in the UV-Vis-NIR, including the optics to focus the laser beam, the mechanical axes to manufacture parts up to 10 cm x 10 cm, specific illumination and equipment for post-curing process. The digital pipeline available will transform the parts design into a 3D printing.

Upgrade & Modification Plan: New extruders will enable the multimaterial robotized printing in case of FFF technology, extending the number of materials that can be processed for one part manufacturing. Hardware and software control will be developed to implement quality control systems during filament manufacturing, and process control based on thermal cameras will be incorporated to the FFF process. For SPP and MPP, the modification optimization of the resins obtained in other BIOMAC-PLs will studied to add the specific photoinitiatior (PI). A system made of a laboratory cuvette, semiautomatic resin dispenser and substrate gripper, synchronized with the laser microfabrication workstation will be set for the SPP experiments. A new set of microscope objectives will be incorporated to SPP/MPP setup for fast fabrication of 3D precise structures.

PL16 (DTI, DK): Printed electronics

The PE pilot line consist of an ink formulation line and an inkjet printing pilot line. The main focus will be to increase automation, throughput and flexibility of each pilot line, in order to bring idea to prototypes below 3 months. The ink formulation pilot lines will consist of rapid screening capabilities, with embedded artificial intelligence, in order to make optimized Design of Experiments (DoE). Print lines will be expanded with high throughput printing increasing flexibility. Bio based TPU and PU resins will be screen printed using ink based on CNC / NFC where viscosity and rheology will be optimized.

Upgrade & Modification Plan: Dispersing and mixing robots will be implemented into the ink formulation screening pilot lines, for full automation. Implementation of AI for fully automated experiments to testing procedures. Upgrading print lines to include a manual dispensing /pick and place system for placing SMDs on flexible substrates using bio-based adhesives developed in BIOMAC.

PL17 (NNT, GR): R2R Nanopatterning and thermoforming.

Topographical nanotexturing and 3D forming by means of nanoimprint & vacuum thermoforming methods on BNC will permit tailored antimicrobial surface functionalisation. NNT PL will be upgraded to accommodate 300 mm wide web PLA films while inspection tools for quality control will be encompassed. NNT micro/nano structuring pilot line will be modified and upgraded to meet the specification of the targeted objectives, such as antimicrobial food packaging materials. The R2R micro-nano structured foils will be converted to sheets with the desired sizes/formats and processed to a post thermoforming step. The replicated and formed micro/nano textured surfaces with complex and highly curved 3D shapes will be assessed to assure the desired properties.

Upgrade & Modification Plan: The synchronisation of all steps within the manufacturing value chain will be adjusted to fit the requirements established in TeC3 food packaging industrial applications. In this context, the fabrication steps for of large area (500 mm x 500 mm) master moulds necessary for the R2R embossing step will be suitable adapted, while the existing R2R imprinting pilot line will be upgraded to accommodate up to 300 mm wide

web films of thickness up to 3 mm. Further to the dimensional (web) upgrade of the R2R pilot line processing variables such as speed, tension, coating, and film surface cleaning/passivation treatment will be adjusted.

Transversal Services: The hub of pilots is accompanied with a series of services that are grouped in three hubs depending on the nature of the service. Next to the PPSH, transversal service mechanism Hubs will be mobilized in order to provide services across the supply and value chain, serving at the same the needs of the TeCs in order to ensure market replication.

B. Validation Services Hub (VSH): This group of services examines the feedstock, the technologies and the products from chemical, environmental, and economical perspective and will consists of:
(i) Quality Control, Characterization: The validation of the novel materials will take place utilizing advanced characterization techniques evaluating functional properties (e.g., mechanical, physical and chemical performance). (PoliMi, UBU) Promoting the bridging of materials developments with end users and real market applications, BIOMAC will enable end users in various fields that will facilitate real field tests of novel materials.
(ii) Standardization: Compliance with standards of all results and defining the specific actions to direct the development activities toward existing standards and regulatory frameworks for the adoption of new technologies by the industry. Proposition of new standards for the nano bio-based materials industry will arise based on the results of the project. (EUBIA)
(iii) Process Validation: Modelling: In order to optimize the reliability of complex systems in the micro/nano range and their performance in real conditions. Simulation and modelling are powerful universal tools by providing suitable failure criteria and models for the evaluation of the reliability of real components. Experimental results from Characterization subsections and the combination of methods will serve to validate the simulations. (IDE, ISQ, IRIS, ABIS, STAM)

C. Market Uptake Hub (MUH): The second group of services is structured to support business, legalization and data handling issues:
(i) *Innovation Management:* These activities will ensure the effective deployment of the exploitation plan provided by the clients of the ecosystem. (EXE) The innovation margins of each result will be identified in market terms. (UNIPD) BIOMAC will provide a variety of innovation management tools that will ensure the prerequisite actions to be taken in order to reinforce the Exploitation Strategy with qualitative and quantitative measures. Innovation management will work in parallel with all other activities of the ecosystem and will indicate the customer segments and which value propositions they require/need. (EXE)
(ii) *Health and safety*: A safe-by-design approach will be applied across the value chains to reduce potential health, safety and environmental risks at an early phase in the innovation process, allowing the identification and management of uncertainties and potential risks Health & Safety (H&S). (UBU) Of special importance is also the use of nanomaterials which should comply to Nanosafety issues in terms of handling however as well as on release from the biopolymer's matrix. (ISQ, UBU)
(iii) Regulation: an assessment of regulation to ensure partnering in the commercialization phase of a technologies, securing compatibility and reducing market uncertainties. (EUBIA, EUBP)
(iv) *Data Management* will carry out the Data governance, Analysis and Design, Database Management, Security, and quality assurance. (RDC, IDE)

D. Value Chain Assessment Hub (VCH): The third group of services is dedicated to the assessment of the value chain:
(i) *Sustainability Assessment:* To demonstrate the final processes and products' environmental and economic sustainability, S-LCA, LCA and LCC will be performed on the develop processes and resulting materials to highlight the environmental impact versus petroleum-based benchmarks. (UBU, ISQ)
(ii) Supply management: Further to this, to enhance the assessment of the overall value chain, a Decision Support Tool (AXIA, RDC) will be developed to simultaneously assess the products against environmental and economic criteria (optimization model). The tool will be further developed in the form of an online user-friendly platform to secure the use from non-optimization experts.
(iii) *Circular Economy:* To stimulate further implementation towards the direction of circular economy, a circular overview of the value chain will be undertaken that will allow the development of efficient networks identifying opportunities to foster the transition from linear to circular model, evaluating the sustainability and economic feasibility. These means of assessment will confirm the future commercial viability and improvement conveying environmental and circular sourcing in consistency with the Green EU Policy. (PoliMi, AXIA, UNIPD)

1.3.1.2 Validation of the OITB through the TeCs

BIOMAC has been built on the basis of a two-stages validation and proof of operation capacity. Initially, 5 Test Cases (interTeCs mentioned at Table 1.1) will validate the proper operation of the OITB and will provide feedback on their experience (e.g., communication with SEP, workflows development, validity of results, times of delivery

etc). During the second stage an **open call for expression of interest** for industry and particular SMEs **will be launched at Month 30** for intra Test Cases (intraTeCs) accessing the OITB. BIOMAC intends to provide to external companies and stakeholders free of charge access in order to develop their TeCs. Again, the selected companies will provide their feedback and thus, the OITB members will have the opportunity to further improve their services. The below table provides briefly the 5 selected interTeCs (the first validation stage of OITB). These interTeCs have been selected based on their complexity in order to involve all the 17 PLs as well as the supporting services and thus provide a holistic feedback for the operation of the OITB as a whole.

Table 1.1: BIOMAC Test Cases (interTeCs) and nano-enabled bio-based material Value Chains

TeC	***Description and involved PLs***	***TeCs***
TeC1: Automotive (DIAD)	Biomass (MISC, EUBIA). Bio-based resins (LTU, BBEPP, AUTH, ATB, FH) reinforced with NFC (LIST). Purification/separation mixed polyols/diols product streams will be also conducted in PL6. In parallel, the PL7 of ATB will produce succinic acid (SA) monomers via fermentation of glucose. SA which will be used in PL12 FH-WKI to produce succinate esters oligomers via reaction of SA with the polyols/diols provided by PL3 (AUTH). UV-curing materials for AM (AIMEN). Properties to be achieved high Temp. stability, low weight, and high ductility for interior applications. (PL1, PL3, PL6, PL7, PL10, PL12, PL15). The polymer matrix will consist of UV-curable Polyurethane (PU) resins fully suitable for additive manufacturing (AM).	
	Unique Selling Points of TeC1 • Biomass/sugar-derived LA and SA reacting with the polyol/diol monomers provided by hydrogenation of sugars, towards highly reactive oligomers that will serve as the basis for the production of PU resins, with exceptional physical properties including toughness, flexibility, and resistance to abrasion. • PU resins, exhibiting elevated temperature resistance (30 °C to 105 °C; 90%+ RH for interiors) • NFC as reactive reinforcing towards enhanced dispersion and interfacial interaction with the resins. optimizing the NFC loading of around 3-5 wt.%. • 15% longer product lifetime and UV resistance through the PU Matrix • Biochar increasing 30% the UV and fire resistance, as well as 20% the thermal conductivity (ASTM E 119) • Fast cycle times and manufacturing of complex geometries (rapid AM prototyping) • Increase and improve the quality of composite structures, reduce rework and wastages, reduce lead-time etc.	
TeC2: Agricultural applications (NVMT)	Biomass (MISC, EUBIA). Succinic acid as a bio-monomer (ATB PL7, AUTH PL3, BBEPP PL5&6) will be used as a monomer of the development of biopolymers. NFC (ITENE PL13) and Nanolignin (Cnano PL4) will be incorporated in the formulation of bioplastic materials biodegradable in soil lending improved mechanical and barrier properties during their use and fertile carbon to the soil in their after-life. Nanobiochar (Uedin, PL8) will be validated as an additive in the formulations to enhance optical properties and UV/thermal resistance while helping soil quality enhancement following bioplastic biodegradation in soil. AIM will produce in PL11 the nanoreinforced PLA pellets. (PL3, PL5, PL6, PL7, PL8, PL11, PL13, PL14)	
	Unique Selling Points of TeC2 • 100% biodegradable in soil according to standards such as EN17033 standard contributing to reduction of microplastics dispersion in soil (white pollution) • Nanolignin (1-3%) incorporated in the bioplastics formulation, provides antioxidant properties to soil biodegradable mulch films and rigid articles used in biological agriculture in substitution to synthetic antioxidant additives • Biochar produced via lignin pyrolysis that will be used as additive in the bioplastics in order to contribute to soil amendment and remediation after biodegradation in the soil • Nanocellulose will contribute to enhanced mechanical strength (ASTM D638) of the bio-based polymers as well as to decreased water vapor transmission rates of agricultural applications (ASTM F1249)	
TeC3: Food packaging (EVERSIA)	Biomass (MISC, EUBIA) surface nanotexturing employed for tailored nano-structuring. (NNT) Vacuum thermoforming will be post-utilized for the production of food containers, bio-based PLA (LUL, BBEPP, AIMPLAS, ATB) foils reinforced with NL and BNC (Cnano, RISE, PoliMi, LIST) will pass the thermoforming PL of NNT. PLA & NFC based inks and coating will be formulated by ITENE (PL14), the barrier and mechanical properties of flexible packaging materials. (PL1, PL2, PL4, PL6, PL11, PL14, PL17)	

	Unique Selling Points of TeC3 • Film made of PLA, Biocompostable and Biodegradable Reel, transparent colour. This product will be produced with material 100% biodegradable and compostable according to EN 13432 standard. • BNC to reach excellent oxygen barrier, high sealing strength, high tear resistance, high stiffness, exceptional surface gloss, high transparency, fully compostable. • NL to increase its storage and life cycle 15%, to maintain life of biodegradable-compostable products • Nanoimprint Lithography (NIL) to increase 20% the self-cleaning and antibacterial activity. >90% increase in bacteria repellence, Water contact angle > 160, Hysteresis angle <10° • Thickness based ISO 4593:2010, µm 20-120, Width, mm 100-1500, ISO 7765-1: 1988 g > 50 • Oxygen Permeability, ASTM D3985, cc/m^2 x 24h <1 • Water Vapour Permeability, ASTM F1249 (38°C & 90%rh), g/m^2 x 24h <5
TeC4: Construction (ACC)	Biomass (MISC, EUBIA). Footbridge module 2m x 0.45 mx 0.65 m, biomaterial functionalities are UV and fire resistance >80ºC, anti-fouling & easy-cleaning capability. Manufactured by multimaterial AM (AIMEN) modified by NFC and Biochar (UEdin, LIST) embedded in thermoplastic isocyanate free TPU biopolymers (BBEPP, ATB, LIST). BIOMAC technology, as an alternate to bridges build using conventional technology and materials like steel and concrete, with low cost, easy of transportation, fast to install, light weight, high strength, corrosion resistant & defect free properties (PL5, PL7, PL8, PL9, PL10, PL13, PL15).
	Unique Selling Points of TeC4 • Production of more sustainable composites (circular economy model) providing a competitive edge and helping to meet future legislative demands and reduce manufacturing cost of composites structures. • Build large all-composite modular structures and minimize the life cost (installation and maintenance). • Enhance safety during manufacturing and installation of large composite structures. • Biochar increasing 30% the UV and fire resistance, as well as the thermal conductivity which are key attributes towards the bridge module to be produced. • Mechanical performance: a) Flexural Strength ≥ 100 MPa, b) Flexural Modulus ≥ 8GPa, c) Bearing strength ≥ 15 MPa (ASTM D790 and ASTM E238) • Thermal and fire resistance performance: a) Tg ≥ 80ºC, b) Reaction to fire Class B ASTM D648, ISO 75. • Thermal conductivity: At least 15-20% lower than current FRP composite 0.3 w/m K at 25ºC, ASTM C51817
TeC5: Printed Electronics (OHMA)	Biomass (MISC, EUBIA). Bio-based inks will be developed reinforced by functionalized CNC (ITENE) and BNC (RISE) using bioresins (LTU, AUTH, FH). TPU produced by REX (LIST) will be used and flexible substrate and nano copper for stretchable electrical prints integrated directly into textiles. (DTI). Bio-based conductive ink for printing onto textiles as stretchable circuitry between sensors and electronics (for stretch and pressure sensing). Conductive bio-based which can withstand up to 50-80% stretching in the fabric will be thus developed. The sock will last at least 3 months with washing every other day. (PL1, PL2, PL3, PL8, PL9 PL12, PL16).
	Unique Selling Points of TeC5 • The overall specification of the materials will be modified with which bio-based polymers and nano-additives on the basis to withstand up to 80% searchability, similar elastic properties as the textiles, sheet resistance below 10 Ohm sq. as well as washability (50 wash cycles) • Final conductive material specs: (Stretchability up to 80%, without damage, Similar elastic properties to textile) • Screen printing ink based on CNC / NFC to protect printed circuits from water and other ambient conditions. • Ink specs: (Screen printable, Viscosity 10-30 Pa s, Dry temp 100ºC, Cure/sintering temp below 150ºC • Final conductive material specs: (Stretchability up to 80%, without damage, Similar elastic properties to textile, Sheet resistance below 10 Ohm sq. Strain sensitive (not essential)

Clustering Activities

BIOMAC will empower the cooperation with other existing test beds NMBP-01-2018, NMBP-03-2019 & NMBP-07-2018 and initiatives in order to benefit from existing knowledge in the framework of competitive models. [12] Clustering activities can be generally described as a group of interconnected organizations that compete and cooperate within the same market and are thus facing similar challenges and opportunities. In relation to the BIOMAC ecosystem, clustering activities will take place on three levels, with (i) research and innovation SMEs (ii) European test beds, and (iii) European or international clusters.

i) The test bed ecosystem seeks not to compete with companies already providing similar services, especially innovative SMEs, but rather cooperate and establish new value chains. Thus, the BIOMAC consortium will establish communications with research and innovative SMEs offering services in the fields of cross sectorial industries in

[12] Shahla G.et. all (2012) Knowledge Sharing in Cross-Functional Teams: A Coopetitive Model, J. Knowledge Management, 16:285-301.

general. This communication will aim at addressing common problems and sharing best practices between the ecosystem and SMEs.

(ii) Cooperation between the test beds funded under the Foundations for Tomorrow's Industry call for proposals (H2020-NMBP-TO-IND-2018-2020) will be a critical aspect for the economic and technological advancement of Europe and the achievement of the goals set forth by the call. Various partners are already involved at the core shell of the test beds funded in the last years and will become liaison to collaboration (e.g. IRIS, AIMPLAS, FH, RISE, PoliMi, EXE, ITENE, AXIA). This will be achieved by scheduling regular meetings between the coordinators, meetings where all participants of the test beds may participate, and organization of conferences and exhibitions where test beds may disseminate their results and attract new customers. Further collaboration will allow the test beds to expand their services portfolio through cooperation. Competition between the test beds will be avoided on the basis of clear business plans and IPR strategies.

(iii) Finally, the ecosystem will establish communications with European and international clusters, such as standardization and scientific organizations (e.g., CEN, ISO, AISC, EMMC, EMCC, EuMat, the Nano-safety cluster). Communications will aim to enhance dissemination of novel technologies and practices, stakeholder awareness, and advancement of standards in the relevant industries.

1.3.2 Positioning of main BIOMAC activities and relation with R&D&I activities

Partners providing the PLs have vast experience in biomass processing and valorisation as well as in nano and biomaterials production (TRL4-5), and they will expand their business portfolio in terms of moving towards a more holistic approach that includes the upgrades summarized beforehand, intending to deliver new supply and value chain business models. The proper operation of the ecosystem will lead to a holistic demonstration reaching a TRL7. Transversal monitoring techniques that have been validated in laboratory settings (TRL4) will be tested in workflows designed to address TeCs. New and adapted test methods will be developed in parallel with the requirements resulting from the different materials (nano scale accuracy needed for the materials involved). The BIOMAC ecosystem will be established with the collaboration of the involved entities, and organisational aspects will be determined for its sustainable operation. Attracting industrial stakeholders will become invaluable due to future replication and market penetration. Conferences, workshops, fairs, seminars, and clustering events will be organized to target a wide stakeholders' audience this will be further facilitated by EUBIA and EUBP. An initial patent search has been conducted by the consortium and no overlaps with the technologies have been noticed. Pilot Line holders' own patents related to mainly processes providing the freedom to operate. The most important patents surrounding the nano bio-based material sector are: EP13001565.2013, ES2398274 B1, ES2398274, EP 2631061 A1.

Table 1.2: *Current and Target TRL of BIOMAC technologies.*

Technology	*Activities for increasing the TRL*	*Final TRL*
Technologies developed in the PLs Hub	Partners leading the PLs have vast experience and processes validated at TRL4-5 covering the overall value from the Biomass Fractionation & Pre-treatment Cluster – BFP through the production of nanomaterial and Intermediate Material and Nanocomposite Cluster – IMNC to Final Products & Formulation Cluster - FPFC that will lead to an actual system development and demonstration. All PL owners have the most experienced personnel with a very big number of publications (Section 4) merged together, where value chain collaborations will lead to the delivery of the holistic approach. See upgrade scheme in 1.3.1.1.	7
Innovation Ecosystem	The BIOMAC ecosystem will be established with the collaboration of the involved entities, and organisational aspects will be determined for its sustainable operation. IBB is a leading network and service company that interacts with industry, academia and politics in the field of sustainable economy in Germany. Established 14 years ago, IBB offers support on technology transfer in the fields of industrial biotechnology and sustainable economy, and has the right position in the sector to act as an SEP managing during as well as post-project this Ecosystem Business opportunity on behalf of the 28 Core members of BIOMAC, strongly supported by Innovation experts of AXIA, EXE and UNIPD. After the PLs are modified, the ecosystem will be validated through the selected TCs reaching a TRL 7 and operate after the finalization of the project in a self-sustainable way post project.	7
Modelling	All PLs are fully operative and have been verified at TRL4 producing the according products thought modelling they will further validate and upgrade their target products and operations. UBU, ISQ and IDENER have been involved in R&D projects acquiring a huge knowledge in materials and process modelling and process optimization. They will correlate their background knowledge and serve test cases to better identify crucial performance parameters and process conditions IRIS also provides tailored artificial intelligence models to use the data from the process monitoring to optimize the process.	7

Process monitoring/ Control	IRIS commercializes NIR-based systems especially targeted at inline monitoring, with high applicability to organic materials such as biomass and biopolymers, but also customized systems working on with ISQ in LightMe OITB project for the automation and process optimization of 6 PLs for the production of nanocomposite materials. Abis has similar experience in Hypersol (FP7) and Finesol (H2020) where they have built a complex high-pressure atomization reactor combined with Process monitoring/Control. The partner's role in BIOMAC will be to develop innovative solutions for monitoring and automation of the project PLs. STAM has a vast experience from PL upgrades from their already involvement in the LightMe and LightCoce OITB.	7
DST	AXIA has experience in the development of DST capable of assessing circular paths building sustainable distribution networks. Similar models are developed for plastic recycling (MultiCycle H2020) and algal biorefineries (Profuture H2020) that will work as a basis for the biomass value chains of BIOMAC.	7

Most of the partners are already involved in relevant National or European projects. The following table describes other R&D initiative related to BIOMAC where for running one synergy with these initiatives will be developed.

Table 1.3: *National and International projects correlating with BIOMAC.*

Projects	***Correlation with*** *BIOMAC*
LightMe OITB LightCoce OITB LEE-BED OITB OASIS OITB NMBP-01-2019	*Most of the partners have collaborated in running or completed projects with* ***POLIMI, IRIS, IDE, ISQ, DTI, ITENE, IDE, AIMEN, AXIA, EXE, Cnano, RDC, STAM, ACC*** *they share common ground and experience towards the development of OITBs.* ***DTI*** *leads and* ***ITEN, AXIA & ACC*** *participate in LEE-BED Test Bed for development and production of nanomaterials for lightweight embedded electronics in sectors such as packaging and automotive.*
EXPEDA National	**AUTH** coordinated bio-based polypropylene/surface treated SiO_2 and CNT nanocomposites and poly(butylene succinate) hemp nanofibers for the development of bionanocomposites.
AQUACOMBINE, RUR-10-2019	The semi-continuous reactor of **LTU** will be used to fractionate halophyte (aquatic) biomass in this project a similar method will be deployed as a baseline.
GOLdJET FUEL, Swedish Agency	The organosolv reactor of **LTU** is used to fractionate wood sawdust with the aim to deliver high quality lignin to be used as feedstock for the production of aviation fuels.
OptiNanoPro, (H2020, NMP-)	**IRIS** and **ITENE** work on PLs in the fields of nanenhanced bio-based materials for injected and laminated packaging, automotive and OPVs in terms of the control systems.
AgriMax (H2020-BBI, ID 720719)	**IRIS, ITENE, AIMPLAS**: Set up of 2 large PL based on flexible multi-feedstocks biorefinery processing, enhancement of NC fibres of NIR systems for monitoring biomass composition.
BIONTOP (H2020-837761)	**AIMPLAS** coordinates and collaborates with **IRIS** and **EUBP** addressing packaging films and textiles with tailored end of life based on bio-based copolymers and PLA through REX.
PERCAL (H2020 -BBI, 745828)	**AIMPLAS** and **IRIS** in PERCAL develop chemical building blocks from versatile biorefinery. PLA and polyols from SA are addressed. Rex is one of the technologies also applied.
APPOLO FP7-FOF-2012 609355	**NNT** was involved in the Hub of Application Laboratories for Equipment Assessment in Laser Based Manufacturing. Providing micro/nano molds for targeted applications (self-cleaning).
NanoUptake (ID 86219, H2020)	**ITENE** leads Computing infrastructure for testing and implementation of safe-by-design approaches in nanotechnology supply chains matching BIOMAC packaging, automotive & AM.
FIBEREUSE (ID H2020-730323-1)	**POLIMI** in the FIBEREUSE project developing bio-based thermoplastic and thermosetting resins for use in fiber reinforced polymer composites improving circular economy approach.
SALTGAE (H2020-WATER-2015, 689785)	**POLIMI** was involved in the development and characterization of bio-based membranes and coatings incorporating algae obtained from saline wastewater treatment plants, for use in the field of composite and nanocomposite materials.
NEWPACK (ID 792261, H2020)	**BBEPP** works on validation in industrial setting PHB- PLA blends with improved sustainability performance, natural extracts with antioxidant/antibacterial properties and nano additives.
AFTERLIFE (ID 745737, BBI)	**IDENER** leads the development of the unified mathematical model regarding the experimental data of the plants, to optimise the operational conditions of the process. **BBEPP** as a partner improves the recovery step by developing a cost-effective process for the purification of PHAs from fermentation broths to be competitive as a bioplastic in some applications.
3DPrintInn: m-era-net	**LIST** coordinate and contribute to this project aiming to develop UV-3D printable fully bio-based inks using components (nanocellulose, lignin, hemicelluloses) from non-woody biomass.
CatBiose: INTER ANR-FNR	**LIST** develops biopolymers based on renewable building blocks from catalytic deoxygenation of low MW hemicelluloses recovered from black liquor for linear and branched polyesters.
GreenCarbon (I721991, H2020)	**UEdin** leads development of new biochar and composite materials and their sequential use in engineering and environmental applications.

1.3.3 Consideration of gender aspects

The BIOMAC consortium supports the objectives of the European Commission to promote gender equality in research and innovation, as outlined in Article 16 of Horizon 2020. [13] Women' under-representation in science due to the lack of attention to gender aspects constitute a major concern in the EU as stated within the Commission's Action Plan on Science and Society, adopted in 2001 (last modification at 2013). [14] Gender aspects have been carefully considered within the consortium in order to ascertain that the promotion of gender equality is present in as many aspects of the project as possible. Equal opportunities are provisioned for all project activities. In this sense, BIOMAC has promoted a series of actions to promote the participation of women and to provide equal opportunities based on gender equality or positive discrimination. [15]

1.3.4 Overall approach and methodology

The project will have a duration of 48 months and will comprise of ten work packages (WPs). Overall, the project consists of four phases: pre-population, population, strategic partnership, and growth. During the pre-population phase (WP1 and WP2), collaboration between all the Hubs will be established to develop the first concepts and TeCs and define requirements for services offered as OSS. WP2 will deal with the upgrades of the PL developing thus the necessary infrastructure for the implementation of WP3 "Pilot Plant Supreme Hubs Services". In WP3, existing inter TeCs will be upscaled in order to be validated in operational environment reaching TRL7. In parallel to this, the other three WPs will start their operation where Validation (WP4), Market Uptake and Value chain Hub services will be developed. Following, the population phase (WP 3-4) will create an initial portfolio of pilot clusters and innovation services in order to support the ecosystem. Below, a summarizing table presenting the Value Chains and collaboration scheme of the OITB in order to deliver TRL7 demonstrators is presented.

Partner	Role	PL (Partner, Country)	TeC1: Automotive (DIAD)	TeC2: Agricultural applications (NOVAM)	TeC3: Food packaging (EVERSIA)	TeC4: Construction (ACCIONA)	TeC5: PE (OHMATEX)
LTU	Biomass organosolv-steam explostion pretreatment	PL1 (LTU, SE)	X		X		X
RISE	Hydrolysis of fibre sludge and BNC production	PL2 (RISE, SE)			X		X
AUTH	Hydrogenation/hydrogenolysis, of sugars towards polyols	PL3 (AUTH, GR)	X	X			X
CNano	US assisted reactor for NL	PL4 (Cnano, GR)		X	X		
BBEPP	Pre-treatment of biomass	PL5 (BBEPP, BE)		X		X	
BBEPP	Recovery / Seperation PL	PL6: (BBEPP, BE)	X	X	X		
ATB	Enzymatic Hydrolysis & Fermentation for SA and LA	PL7 (ATB, DE)	X	X	X		
Uedin	Pyrolysis of Lignin (Biochar)	PL8 (UEdin, UK)		X		X	X
LIST	Reactive Extrusion for TPU	PL9 (LIST, LU)				X	X
LIST	Cellulose mechanical treatment	PL10 (LIST, LU)	X			X	
AIMP	Reactive extrusion for PLA	PL11 (AIMPLAS, ES)		X	X		
FH-WKI	Polymer resins polyester and PU resins	PL12 (FH-WKI, DE)	X				X
ITENE	Cellulose mechanical treatment	PL13 (ITENE, ES)		X		X	
ITENE	Nanocellulose based coatings	PL14 (ITENE, ES)		X	X		X
AIMEN	Additive Manufacruting	PL15 (AIMEN, ES)	X			X	
DTI	Bio based Printed electronics	PL16 (DTI, DK)					X
NNT	R2R NIL Thermoforming	PL17 (NNT, GR)			X		

Figure 1.3: *BIOMAC Value Chains immobilization*

The strategic partnership phase (WP 5-7) aims to continue the growth of the ecosystem, further than the existing network. This phase will also focus on creating additional circular business models and appropriate strategies that will bring post project revenue streams to the ecosystem. Finally, the growth phase (WP 8-9) is intended to create substantial returns to the ecosystem. This phase will be expanded post-project, proving BIOMACs sustainability. Thus, in parallel to all WP developing services (WP3-6), IBB in collaboration with EXE and UNIPD will start with a strategic way to build its post-project sustainability where key tools for the proper operation will be organized, established and improved throughout the entire duration of the project (WP1, WP8, WP9). In M20 starting WP7 tenders for the open call will launch selecting external intra TeCs (M30) that will use the OITB and the ecosystem

[13] DIRECTIVE 2004/113/EC
[14] SEC (2001)771, "Women and science, the gender dimension as a leverage for reforming science"
[15] http://ec.europa.eu/programmes/horizon2020/en/h2020-section/promoting-gender-equality-research-and-innovation

services OITBs. In parallel, these cases will be examined from the business and innovation perspective (WP8 & WP9). All necessary activities for sustainability (WP8), dissemination, communication, and clustering (WP9) have been projected. All this will be governed under coherent managerial and organizational structure (WP10).

1.4 Ambition

1.4.1. Advancement beyond the state of the art

The overall ambition of BIOMAC is to boost and sustain innovation in the field of European bio-economy industries, by reducing the time-to-market of novel nanotechnologies, thus reducing costs and risks. The BIOMAC ecosystem will act in a sustainable way by the core group of the ecosystem (PL owners and service providers) after the implementation of the project. The exploitation strategy and dissemination actions foreseen will ensure that potential customers will be informed about the existence of the test bed, services, and access conditions.

Innovation #1 - Enhanced and new material functionalities through the production and use of nanomaterials: BIOMAC involves the complete value chain, including: pre-treatment and conversion of biomass, biopolymer production pilot lines, process monitoring and scale up industrialization services. The objective of BIOMAC is the production of nanomaterials from biomass through a holistic and integrated approach by maximizing the use all components and the improvement of conversion technologies, and finally the integration of nanomaterials across the manufacturing through the development of reliable and sustainable production of functional nano-enabled biomaterials. Linking properties and process conditions with the quality of the final product is of utmost importance and the basis for accurate process control. While for SMEs and innovative industries it is often unclear which technologies might be most suitable for later process monitoring due to budget constraints and business deviation, the 17 PLs of the Ecosystem will be ready to be customized and accessible to the Ecosystem's clients. Thus, clients of the Ecosystem will come in contact with multiple or single step processes, based on a holistic supply chain management approach.

Innovation #2 - A single entry point ecosystem for nano based biomaterials

Already from the first month of the project the SEP, represented by IBB, will be organized. The SEP will be coordinated by the Management Board (MB) where each one of the 28 core partners will have one vote. It should be mentioned that all entities will involve decision makers who have the capacity to represent their business. All necessary legal documents (NDAs, contracts etc) will be developed. The Business and Exploitation Plan of the BIOMAC SEP will include, mission and Vision, the governance structure, the selection procedure of the BIOMAC representative and additional articles as it will be agreed between the partners. The Management Board will be also responsible to monitor the progress of the SEP and the performance of the general manager based on achieving quantifiable Key Performance Indicators (KPIs). The structure and the interactions between the 4 different Hubs of the ecosystem is shown in (Figure 1.4). The MB of the BIOMAC ecosystem will include the following roles: (a) the Ecosystem Manager that will be the President of the MB, (b) the Innovation Manager, and (c) the Pilot Hubs Manager that will be supported by the Data Manager, the Regulatory and Safety Manager, the Validation Manager, and the Value Chain Manager. The persons that will be cover this role will be elected by the MB for a service of 3 years. The MB will be in charge of defining the ecosystem strategy, including exploitation plans, the IPRs, the pricing of services, the data handling, standardization and Safety Assessment, and the marketing activities of the ecosystem. The MB will also ensure the cooperation among other existing related ecosystems and Innovation Communities with an aim to make the cooperation systematic and sustainable at the end of the project. The MB will also ensure the cooperation among other existing related clusters and will also communicate with an Advisory board (AB consisting of well-established entities in the field of nanotechnology and metal processing industry).

Figure 1.4: *Organisational and operational structure of SEP*

Innovation #3 - Ecosystem advantages: **Enhanced performance:** BIOMAC will target material modifications at the nanoscale, promising to deliver the optimization of material behaviour and performance needed to significantly improve the mechanical performance, durability, multifunctionality and sustainability, while resulting to nano modified biopolymer structures and products. **Fast screening**: Through the database that will be developed under BIOMAC, based on the TeCs and new clients brought in the Ecosystem, fast screening of TeCs will be achieved. **Fast characterisation:** Predictive modelling of materials, (IDE) elements and components within BIOMAC will aim to

optimize products based on market requirements and to ensure their proper functionality. **Expertise:** In particular, tacit knowledge will help organizational activities and functions by creating new knowledge, targeted to new product development, novel business concepts, and procedures. **Recognise next actors**: The following steps will be followed: i. identification of technology change and contact relevant stakeholders, ii. taking into consideration research insights, iii. following changes in customer demand, iv. identify competitors, v. consideration of policies and legal environment.

Innovation #4 - Establishment of an open innovation ecosystem in the biomaterials field: The BIOMAC ecosystem will form a pan-European network of experts in the field of biomaterials and will provide open access to **17 PLs**, as well as complementary services along the entire value chain, meeting the needs of industry. The network will be based on the open innovation philosophy and thus, inside the Ecosystem knowledge transfer and efficient combination of resources will take place. The network will outreach new users' "clients" across different regions of the EU, sharing best practices and experiences, identifying new innovative technologies that can benefit, brokering between users and suppliers, leveraging further investment by creating linkages with regional/national initiatives and by stimulating business growth. Increased profitability will result from the collaboration of experts in different technology and service fields, while a shortened time-to-market, especially for SMEs that have a lack of resources will reduce their financial asset commitment and enhance their flexibility. Moreover, through transferring their know-how and resources, the participants will be able to expand their knowledge bases and competences, enhancing innovation and learning. To this intend EUBP will also decisively contribute being the largest association and network within the European bioplastics industry, drawing an extensive close-knit network of key personnel in organisations involved in bioplastics research, sourcing, production, distribution, consumption, as well as stakeholders (BASF, DuPont, Ferrero, Lavazza, Mitsubishi Chemical Europe and many more) contacts involved in the pertinent political processes, associations, NGOs, and civil society organisations, through this extensive stakeholders network and the high amount of contacts the OITB will reach its fullest potential.

Innovation #5 - Integrated Innovation Management: The BIOMAC ecosystem business model will use a Commercial driven impact model addressing its client's needs. This model comprises 3 synergistic tasks to help accelerate economic outputs , namely– **a)** IP protection (patent filings at EUPO and USPTO) to protect the strategic and commercial sensitive innovation thus keeping competitive advantages "within Europe" and giving the ecosystem's clients an important competitive advantage on a global level, **b)** business and go-to-market plans comprising key enablement essentials such as market feasibility, opportunity analysis, client segmentation, risk analyses, ultimately delivering a fully substantiated business plan, and **c)** a dedicated Platform as part of pan-European Network Innovation Networks in order to maximise the impact of the project, incorporating domain relevant commercial actors, prospective industry partners, international contacts and VCs and/or Angel investors. This new Platform will be initiated through the Innovation Strategy Forum to include, beyond the consortium members, over domain-relevant stakeholders, thus acting as an acceleration vehicle towards incentivizing commercial and economic outputs from the project. BIOMAC will also reach out to previously funded H2020 and BBI projects including OITB for EC investment multiplication and will link up with relevant networks.

Innovation #6 – Sustainability of Value chain and EoL management: The BIOMAC ecosystem will build the business model and the solid foundation of the circular economy to maintain an effective plan towards sustainable reuse and recycling practices that will allow the demonstration of effective and suitable per case EoL management. Based on the analysis of the sustainability of the value chain, the LCA and LCC assessments, a Decision Support Tool (DST) will be developed in the form of optimization model, materializing all the critical factors (cost, efficiency, environmental restrictions, recyclability) recommending an optimal network of connections supporting the creation of circular processing paths and material's use.

1.4.2 Patent search

The consortium has additionally carried out an extended patent search in available patent libraries (European Patent Office, USPTO, FPO) to specify possible opportunities, not only to achieve knowledge advancements, but also to proceed in technological advancements in terms of patenting. Thus, the patent search was conducted in the spirit of investigating and detecting unexplored areas and approaches of the field and to avoid possible overlapping with existing protected intellectual knowledge. The decision of the consortium is that although a number of patents exist in individual fields covering all processes and materials, there is a lack of holistic approaches across the whole value chain. In the following table the results of the internal consortium patent mapping are presented, and it is evident that the partners will benefit from joining forces and creating vast Value Chains through the demonstration of TRL7 user cases. This would allow to exploit the business lines without infringing any IP as well as giving competitive value to the enabled products.

Table 1.4: Patent mapping of partners

Patents	Overlappings
WO2009018834A1 WO/2004/085529	[AUTH] owns more than 15 international patents in biotechnology and 1 in nanotechnology, "Composite compositions with barrier properties and methods for the preparation thereof".
FR 2854161	[LIST] Crystalline polysaccharide derivatives in the form of water-insoluble aggregates of microcrystals for use as viscosity modifiers or super-absorbers, manufactured by controlled oxidation of primary alcohol groups.
ES2364831	[AIMPLAS] Procedure to obtain thermoplastic starch and thermoplastic starch obtained by this procedure.
EP 2784114 A1	[AIMPLAS] Bio-based and biodegradable polymer to be used in the manufacture of films for different applications.
EP13382126	[ITENE] Composition for the Preparation of a Nanostructured Biodegradable Polymeric. Material, the Material Obtained and its Applications.
EP18198458	[IRIS] integrated optical system for the examination of sample materials.

Additionally, a more detailed and specific technology Landscape was explored, through the licensed PatSnap and Discovery (Market and Competitors) software, and from this preliminary analysis the End User demonstrators are unique, also due to the fact that the PLs are based on rather innovative and upgraded processes, allowing advanced functionalities. The patent search followed a multi levels approach on the sector, starting by available patents using key words as following: Bio-based Polymers → Nanomaterials Nanoparticles → NL, NFC, BNC, CNC. This refinement leads us to the specific market and indicates us the "***freedom to operate***" since by this procedure the presence of very few patents that are mainly filed in China was concluded.

Key Word	Simple	Total
Bio-based Polymers	4.307	13.249
Nano-enabled bio-based materials	532	1.605
Fibrillated nanocellulose	421	1.582
Bio-based Polymers nanocellulose	79	265
Bacteria nanocellulose polymers	70	178
Bio-based Polymers nano lignin	10	36

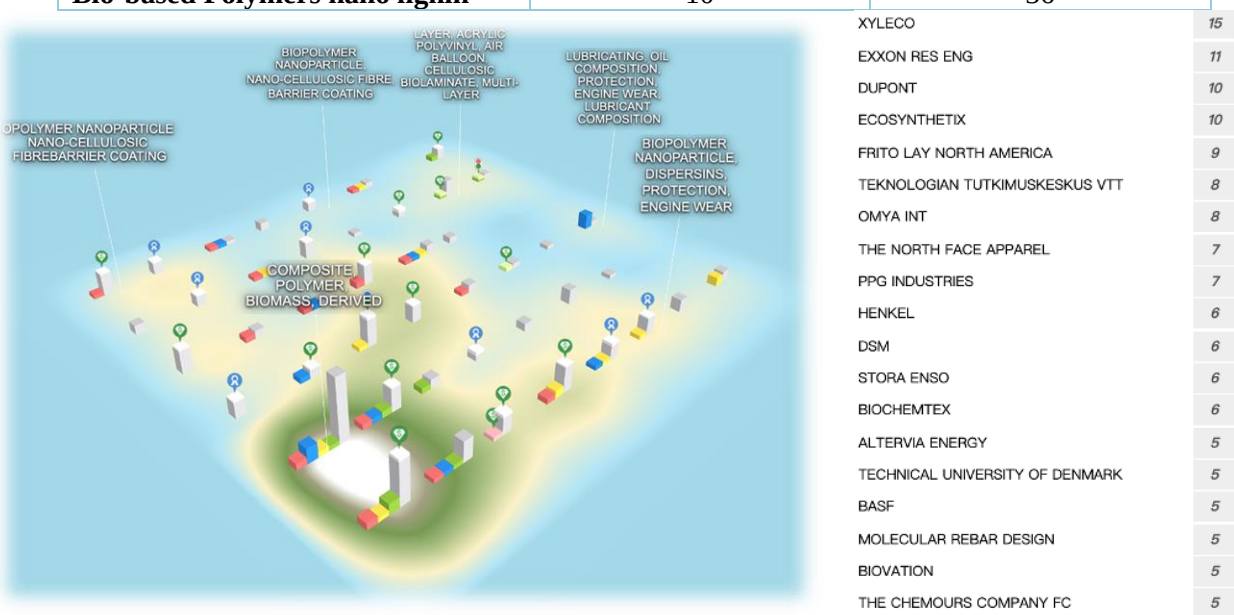


Figure 1.5: *FTO PatSnap (Patents, left) and Discovery (Market players, right) software*

1.4.3 Existing innovation test beds/HUBS

BIOMAC covers the entire value chain from raw materials crossing versatile Pilot Lines reaching out towards the Test Cases, which compliments this with tailored services in order to rationalise concept to pilot production of prototypes and accessing protocols. The Open Innovation environment did its first steps in H2020 and a limited number or similar initiative can be identified, with these that are related to BIOMAC shown in Figure 1.9. In general, all these HUBS are very strong technically or for networking, however they lack the support services foreseen in BIOMAC. BIOMAC will offer a fully holistic approach, providing services and pilot lines that will allow companies to go from idea to prototype to full production. In the table below, some national ecosystems have been found and reported. Although they create business cases and examples which are very promising, BIOMAC is advancing above the state of the art by presenting: (A) Single Entry Point (SEP), (B) Pan–European ecosystem, OSS and (C)

Multidisciplinary approach bridging different Pilots and sectors. In Georgia (US) the **IDEAS:MD3** is based, which is the latest centre in Georgia Tech's continued expansion of its Materials Innovation Ecosystem. Its mission is to develop the initial data engineering and science foundation in that ecosystem, a key to unleashing new potential in solving grand challenges. [16] This ecosystem will become a point of reference which BIOMAC will use as a good example. Again, in the US the **Michigan Translational Research & Commercialization Hub (MTRAC)** consists of an Innovation Hub for Advanced Applied Materials, sponsored by the 21st Century Jobs Fund through the Michigan Strategic Fund and administered by the Michigan Economic Development Corporation (MEDC), to support collaborative translational research projects led by teams of researchers, and business advisors. [17] The mission of the MTRAC Innovation Hub is to advance materials-based technologies that address poorly met market needs into commercial development.

Figure 1.6: Other global test beds and innovation ecosystems

In Switzerland a **Center of Excellence in Characterisation (CSEM)** exists, offering mainly Excellence in Characterization, bringing together experts and methods from CSEM, the Haute Ecole Arc (HE-Arc) and the Paul Scherrer Institute (PSI), in order to meet industrial requirements, offering easy access to the instruments used for materials characterization. [18] BIOMAC will become a pan-European network, thus will seek collaboration benefiting its clients, mainly located and serving the European market through a virtual platform. **Nano for Production** is an innovation hub located in Fraunhofer (FH), Germany, it is adjusting structures smaller than 100 nm using them in industrial applications. BIOMAC is complementary since its industrial scope is to use and utilize nanomaterials in new nano-enabled bio-based materials end user applications, and BIOMAC will seek links with the innovation hub of FH during and after its operation.

1.4.4 BIOMAC Value Chain position in the market

Today, there are multiple types of biopolymers based on different raw materials, and new products or processes across the Value Chain are regularly appearing. However, joint marketing activities by manufacturers, the development of new biopolymer materials and products, and the increased availability of bio-based products will contribute to promote new Supply and Value Chains as it has been planned in BIOMAC. BIOMAC, creating an Open Innovation Test Bed intends to reach Technological advances as well as to raise social messages at Regional and European Level underlining the importance of sustainable development providing the bioplastics industry a holistic materials OSS approach that would lead to highest innovation rates, faster prototyping and lower the market entry time. Considering that biopolymer prices are also expected to fall in coming years, as economies of scale, increased experience levels as well as technological innovations help to lower the costs of producing and processing them, BIOMAC identifies a huge market as it will be analysed in the Business Plan of Section 2. The increase in prices of fossil resources will also contribute to reducing the price differentials. Through the wide and interdisciplinary network of BIOMAC team members, the OITB will provide access to technological infrastructures and expertise of End User industries to produce innovations and deliver nano-enabled bio-based materials. It also will offer business support for industries to implement digital technologies, where needed across their value chain, and reduce discrepancies between industrial sectors. EUBP, UNIPD and EXE are skilful in working with exploitable results from technological research projects and can design business plans for ensuring market acceptance and long-term viability of technological solutions. BIOMAC will deliver costumer-based sustainability plans for its clients and undertake the implementation of feasibility studies for novel market strategies. The aim is to develop economic self-sufficiency business models understanding the dynamics of business development and innovation. **Prices** for **biopolymers** vary and will be used as baseline, starch polymers **cost** from €1/kg; starch blends, €3-3.50/kg; cellulose polymers, €3.40-4.50/ kg; and polylactic acid (PLA), €2.30-3.50/kg. For example, DuPont and Tate & Lyle have jointly developed a proprietary fermentation and purification process to produce bio-based 1,3-propanediol (PDO). **BASF produce Ecovio®** a blend of PLA and the fossil based polyester **EcoFlex®.** Starch-PLA blends (e.g., Cereplast **Hybrid™**) are available and PLA/PHA blends are being investigated. DuPont produce two additives specifically designed to enhance the properties of PLA; **Biomax® Strong** enhances impact strength, flexibility and viscosity. **Purac** is world's largest supplier of lactic acid (100,000 tonnes per annum), derived from cane sugar or tapioca starch. **Natureworks** LLC also manufacture lactic acid, from corn starch, and are the world's only large volume producer of PLA (150,000 tn per annum), their PLA costs €1.90 per kg. Demand for PLA is increasing rapidly, and it is expected to be the largest bioplastic produced in the US by 2011 at > 300 million lbs. For these reasons both polymers encounter hurdles which are avoided by using alternative feed stocks and through key PL processes of BIOMAC.

[16] www.materialsdev.gatech.edu/ideas-md3
[17] https://cfe.umich.edu/mtrac/
[18] https://www.csem.ch/Page.aspx?pid=47739

2. Impact

2.0 BIOMAC Impact at the level of operation

BIOMAC promotes a one-stop- shop offering all the services needed to make commercial applications, based on open technological infrastructures, crucial to the technology maturation. It has been decided that the Ecosystem will be represented by **IBB which is a well-known and established network and service company in the field of bioplastics** acting as the **Single-Entry Point (SEP) for the OITB**. As IBB will be acting as SEP, its organizational structure is following to the German association law, providing association autonomy according to § 21 BGB (German Civil Code). As such a non-economic association (so-called ideal association), BIOMAC's SEP unit receives the status of a legal entity and thus becomes a legally responsible corporation. Acting as non-economic association will also ensure cost-reduction for the whole consortium. **IBB** will start from the early stages of the project to work towards the formalities for the BIOMAC OITB and the SEP operation. This includes communication with the clients, set up of special contracts between the SEP and the core partners to have the power to represent the ecosystem by closing agreements and signing contracts with the potential customers, in accordance with national & EC regulations.

1. BIOMAC Value proposition: The BIOMAC Ecosystem will provide open access to the physical facilities (17 Pilot Lines) and services required for the development, testing and upscaling of materials and products in the field of nano-enabled bio-based products and materials. **Open Access** in this context means that any interested user (e.g., SMEs, Industry, Start-ups, Universities, Research Institutes and other stakeholders) from Europe and beyond, can have a **non-discriminative access** to OIT facilities, capabilities and services. under **fair conditions and prices** and with transparent and mutual obligation with regards to, for instance, security and safety of IPRs. The BIOMAC Ecosystem will be **self-sustainable** after the project end and thus its operation will target on the creation of revenues from its different activities. The administrative burden for the customers will be minimum as they will come in contact only with the SEP structure) for setting up the service agreement. The main services that will boost the innovation capacity of the ecosystem are:

(A) technological services: providing access to expertise, technology and infrastructures, including manufacturing and product development. This includes for instance: collaborative RD&I, technical support on scale-up, commercial infrastructure, testing and validation, etc.,

(B) ecosystem services: developing and driving open innovation ecosystems with strong cooperation between research and industry and good connections between users and suppliers of innovations across the bio-based value-chain. This includes for instance: dissemination of information, innovation strategy development, brokerage, inter-collaboration, etc. and

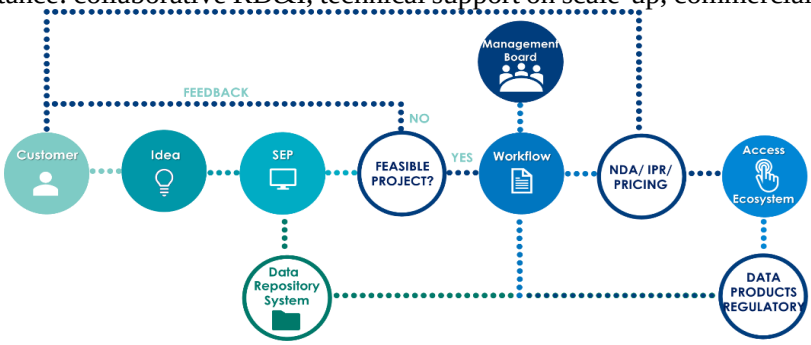


Figure 2.1: SEP Operation & Access

(C) business-oriented services: supporting businesses to commercially apply these innovations and transform their business, including access to financing and investors. This includes for instance: technology transfer, IPR exploitation incubator/accelerator support, access to finance, business/market intelligence, etc. Innovation management in the open innovation environment will enhance information sharing and diffusion of expertise for all actors involved to increase the chances of success, lower risks, and reduce costs.

2. SEP structure, management and procedures:

- ✓ A framework agreement between IBB (SEP) and the other core partners will be defined. Based on this agreement, the offer of each core partner will be defined, while the SEP will be given the mandate to sign contracts on behalf of the OITB with the clients. The SEP will remunerate the involved partner based on their effort for the implementation of each new test case brought by a client.
- ✓ Pricing policy – payments for the services, obligations of the core partners towards the SEP, liability (if any), confidentiality issues and IPRs will govern the multi actor relationship.
- ✓ An agreement template between the SEP and the potential clients covering issues related to the cost of service, access conditions, confidentiality and IPRs.
- ✓ To ensure sustainability of the Ecosystem, the above agreements will be based on the business plans that the core partners have already developed which will constantly be updated.

3. OITB & SEP operation & Advisory Board: The SEP (IBB) will be operating under the guidance of the OITB MB. The MB will define the operational strategy and the relations of the OITB with its client. It is estimated that the SEP will initially need 2-3 employees in order to carry out its tasks. The OITB MB will communicate with an Advisory board (AB) consisting of recognized persons in the field of bio-based technologies and innovation management. The Advisory board members will come from associations, stakeholders, and clusters that are willing to act as advisors and supporting mechanism of the BIOMAC ecosystem. Already *15 entities* confirmed to become members of the AB will be *Giulia Gregori (Cluster SPRING), Dr. Monica Normark (Sekab E-Technology), Ulf Westerberg (Pitea Science Park-Bothnia BioIndustries Cluster), Evangelos Karavas (Pharmathen), George Koliakos*

(Biohellenika), Anastasios Rizopoulos (Interplast), Joseba Bilbatua (MANU-KET), Miguel Angel Rodiel Moros (MATERPLAT), Sergio Sáez (SECPHO), Dr. Antonio Manrique, (CEO KIRAM), Carmine Pagnozzi (ASSOBIOPLASTICHE) which will offer professional advice and expertise on the field of automotive, construction, food packaging and printed electronics. Based on their knowledge and market experience they can contribute towards this end, while having access to the publishable results of the project and through their participation in Ecosystem events. A draft business plan of the Ecosystem is presented in Section 2.2.

2.1 Expected impacts

Realisation of open and upgraded facilities at the EU level for the design, development, testing, safety	
BIOMAC aims to set up a SEP through which a holistic portfolio of versatile PLs and transversal services open to European midcaps and industries will be offered. The BIOMAC will take advantage of upgraded facilities and expertise in various locations across the EU (GR, BE, HR, SE, LU, DE, UK, ES, IT, DK and PO). It is planned to upgrade all PLs at TRL 7 and setup a transversal service mechanism. The OITB and its SEP will work towards an open innovation environment in which the clients will develop, co-develop and share knowledge and expertise with the OITB and thus promoting innovation in the field of bio bases materials. The application for access in the OITB will be done through a special designed web-platform that will be supported by the members of the OITB providing fast and reliable assessment to clients and setting up dedicated workflows.	
KPIs:	• 17 Upgraded Pilot Lines across the value chain (BFPH, IMNPH and FPFH). • 5 selected cases accessing the OITB as first validators (DIAD, NVMT, EVER, ACC and OHM). • 5 additional cases accessing the OITB via an open call.
At least a 20% increase in the number of new SME users for existing test beds;	
BIOMAC will be open to related companies to access at fair conditions and costs. Already during the setup, BIOMAC has sought the market potential through the existing customers list of the RTOs and through SMEs within their innovation cycle. Thus, the network already accumulated a list of multiple industrial stakeholders which expressed interest to become the first potential client users validating it. It has been scheduled to launch an open call the second year of the project targeting an open SEP approach. Effective access to finance and uptake by industry will be facilitated through the digital interface and the dissemination and business exploitation plans. The digital interface will have two operations: (i) a platform of RDC where customers can apply to make use of the ecosystem's services and (ii) a lighthouse, of new technologies, where crowdfunding can be initiated, and technology investors can express interest on entering the OITB. Further SMEs entering BIOMAC will raise additional funding and revenues from services provided to TeCs that will enter by the Open Call. On the other hand, besides the general dissemination activities, specific actions are projected targeting to facilitate the access of SMEs. For example, Task 8.3 is dedicated to the provision of additional funding for possible clients, while the WP07 for the open call will allow the OITB to come in contact with a large number of potential clients.	
KPIs:	• 50 potential clients contacted during proposal preparation phase. • 40 have already provided a letter of intent. • 1,000 industrial partners (~70% of them SMEs) will be reached during the project. • At least 50 stakeholders to become members of the BIOMAC community by the project end.
At least 15% improved industrial process parameters and 20% faster verification of materials performance:	
This impact heavily depends on process validation and cost control, for example pre-treatment offers unique advantages when compared to conventional batch process, such as reduction of the energy consumption up to almost 50%, significant reduction of the lignin content in the cellulose fraction and increased recovery of hemicelluloses and lignin fractions. A process control approach will be developed, which can largely improve the process efficiency, allowing to find complex, non-linear patterns in data, and transform them into models, which can be applied to fine-tune the process parameters, improving them by at least 15%. Process modelling and simulation will firstly help to identify the key parameters of each of the industrial processes involved in the PLs and set the basis for the process optimization, that will be performed in combination with the monitoring and closed-loop control of the process. By means of predictive modelling and simulation, materials targeting highly promising applications will find a preferential route of performance verification through the Validation Services Hub, leading to an at least 20% faster verification process.	
KPIs:	• BIOMAC will reduce bio concepts development by a factor of 2 compared to the current status. • BIOMAC will reduce concept to prototype component to under eight months (e.g., 3M for PE).
At least 20% improvement in industrial productivity, reliability, environmental performance, durability…	
BIOMAC will enable the cost-effective production of high-value nano-enabled biomaterials that can address performance challenges. In addition, further development of energy-efficient, low environmental impact, and lower cost production can have significant impacts within the European bioeconomic sector. An at least 20% improvement in different levels will be achieved due to the tools and equipment used in processing that uses less energy (per unit) for a much shorter duration relative to conventional pre-treatment and synthesis approaches. Additionally, there will be a reduction in embodied energy and GHG emissions for polymers. Finally, process monitoring, combined with fast characterization methods, will increase productivity and equipment lifetime, and	

reduce production costs, increasing production capacity by 20%, while lowering material consumption rates by 4% and downtime by 20%.	
KPIs:	• Eliminate the use of fossil-based monomers thus produce bio-based polymers and derived NPs from all fractions of the biomass to produce BNC, Biochar, NC, NFC and CNC. • Increase of components lifetime by 35% even for the most demanding applications (e.g., construction and automotive), reducing thus maintenance costs. • Decrease raw materials usage by a factor of 25% for both NIL, PE and AM technologies due to the upgrades on the PLs (e.g., in-line monitoring systems, preventing the production of fault batches, re-use of scrap and non-printed material).
Facilitating access to finance (for SMEs in particular) for investing in materials or in applications using…	
Having access to BIOMAC will eliminate the barrier of potential stakeholders (large enterprises, SMEs and midcaps) to realise new ideas accessing new and unexplored markets and creating new ones. The Market Uptake & Innovation Cluster will be established in order to support industry on compliance with regulatory and standardization requirements, since the early stages possible of product and/or process development, to validate the performance and to guarantee both process and product quality and safety. In order to achieve the specific objectives related to "C" that are to secure the operation and sustainability of the ecosystem various activities have been foreseen such as to attaining funding through national/regional, Venture capitalists and business angels, public and private funding sources (Crowd-funding platforms such as FundedByMe (Sweden), Spreds (Belgium), Ulule (France), Booomerang (Denmark) and Seedmatch (Germany), EU research and innovation funds, European Regional Development Fund (ERDF), InnovationRadar, Dealflow.eu, VentureEU, InvestEU, European Investment Fund (EIF) e.t.c.). Recognizing the importance of small and medium-sized enterprises (SMEs) for the European economy, BIOMAC aims to enhancement the traditional corporate strategy perspective with a number of cases, mainly driven by the biomass and biomaterials processing sector. Innovation management strategies based on the identification of technological benefits and thorough business planning are developed on the unique characteristics of each case, with clear understanding and agreement between stakeholders on relevant intellectual property rights. Additionally, BIOMAC will contact investors, interesting to monitor the progress of the technology and the market uptake thus use financial support for pilot projects and their market absorption. Further to this, DIAD, NOVAMONT, EVERSIA, ACC and OHM through BIOMAC, intends to establish some collaboration agreement with SMEs to share IP and costs which would facilitate SMEs access to BIOMAC.	
KPIs:	• Organization of 3 events with participation of investors and funding groups for showcasing of novel concepts by SMEs during the project. • Demonstrate funding of test cases for SMEs during the open call phase using BIOMAC funding service
Identification of regulatory requirements and regulatory barriers to market introduction;	
The European Bioplastics Association (EUBP) is partner of the consortium and will contribute to aspects to regulatory and policy framework for new bioplastics. Special attention will be given to nanomaterials regulations (nano-safety), guidelines and procedures in order to achieve a smoother market introduction by ensuring safety work conditions and utilization of the developed products. Special certifications for Nano-enabled materials can facilitate the choice of the products by the consumers. Regulations and Standards for early detection of possible conflicts related to BIOMAC developments will be done, assisted by existing tools and agencies as the EU Observatory for Nanomaterials (EUON), considering of industry standards (such as EN13432), EU Agency for Safety and Health at Work (EU-OSHA) and OECD.	
KPIs:	• Participation in more than 50 EU consultation & other worldwide regulatory in the field of interest • Transfer of the BIOMAC safety assessment to the standardization committee

2.1.1 Other expected impacts

A. Release of the European Union from dependence on fossil-based materials

The use of fossil-based materials is reflected into a dependence on fossil oil producers and their countries. Indeed, due to limited extraction of fossil oils in the EU, the EU industry (and employment) is strongly depended on extra imports, mainly from Russia, Azerbaijan, Kazakhstan and OPEC. [19] It is well known that some of these countries may be critical for political or social reasons, therefore the price of the fossil-based materials risk not to be stable. Moreover, the countries extracting the fossil oil may produce (or may decide to produce in the future) fossil-based materials (e.g., plastics) at lower cost compared to the EU industries. Therefore, the EU occupation in the sector of fossil-based materials and products risk to be fragile, as it depends on extra EU purchases of raw materials and extra EU industrial strategies. [20] The production of bio-based materials does not require extra EU imports since it is based on regional EU value chains in close cooperation with primary sector and is managed entirely at EU level, given the extensively developed European BBI industry and the subsequent wide availability of feedstock. This may consist in a more stable occupation, product price and material availability.

[19] https://ec.europa.eu/energy/en/topics/imports-and-secure-supplies/supplier-countries

[20] https://www.transportenvironment.org/sites/te/files/publications/2016_07_Briefing_Europe_increasingly_dependent_risky_oil_FINAL_0.pdf

B. Socio-Economic impact – Jobs creation:

The chemical industry is of huge importance to the EC. The European bioeconomy in the food, feed, beverage and primary sectors, has an annual turnover of € 2.1 trillion. Bio-based industries (relevant chemicals, plastics, paper and paper products, forest-based industries, textile sector, biofuels, and bioenergy) account for € 600 bn of this total. Moreover, the European bioeconomy provides jobs for 18.3 million Europeans, over half of them in primary biomass production (agriculture forestry). Bio-based industries employ 3.2 million people. [21] BIOMAC will promote the bio-based value chain that will create a multidisciplinary technology driven partnership strongly contributing to the creation of new jobs in the area of bio-based materials and processing. European industry adopting the new products and procedures will maintain the EU market and claim the global market, based on the knowledge obtained during the project, a fact that will strengthen their competitiveness. Ecosystems need appropriate staffing to become sustainable. The process of opening up and further developing access to existing and new industrial infrastructures requires careful management, investments and high expertise stuff. Integrated successful Ecosystem activities will lead to the creation of job vacancies of high level technical, scientific, managerial, organisational, and marketing personnel in the members of the Ecosystem and the SEP. On the other hand, successful outcome of the test cases accessing the Ecosystem services will lead to a significant growth of the corresponding companies (including SMEs) as they will bring to the market novel and sustainable Bio based nano materials. Being established on the market with innovative nanomaterials will allowing the SMEs to also diversify their revenues, enhancing the exploitation chances. Based on the preliminary business plans carried out for the Ecosystem and the selected test cases it is evident that an important revenue will be created, leading to the creation of new high expertise jobs. More specifically it is projected that 5 years after the conclusion of the project, **380 new jobs**, will be created for the members of the Ecosystem and **63 new jobs** will be for the end users. Taking into account the jobs that will be created on the end users for a period of 5 years after the project end, **the BIOMAC project, will lead to more than ~1650 new jobs in total.**

C. Steps towards meeting the Paris agreement goals

BIOMAC will be developed around using biomass as raw material for nano reinforced plastics production and create a complete supply and value chain around the creation of nano-enabled bio-based materials. After the project completion, these technologies will be applied to biomass that is produced from other sources. Further on, the use of biomass will be evaluated based on the needs for process adjustments and possible additional treatment. Expansion to these sources of biomass will exponentially increase the number of materials that can be used for the production of bio-based plastics in a global scale. These applications will significantly expand the impact on the global economy. BIOMAC will reduce the overall CO_2 footprint from human activities [22] and push the global economy towards renewable resources. For example, the enhancement of the properties of agricultural products biodegradable in soil will contribute to avoid the use of non-biodegradable plastics in agriculture, today the principal cause for soil depletion, in particular by contributing to (i) avoid microplastics dispersion in soil and (ii) restoring carbon in the soil by supplying fertile carbon to improve the Soil Organic Matter. These factors heavily affect the health and quality of the soil which is a key player not only for the productivity of agricultural soils, which is constantly decreasing, but also for the fixation and storage of atmospheric CO_2 into the soil.

D. Increased growth of Start-ups and SMEs

BIOMAC will ultimately contribute to: (A) encourage owners of Pilot Lines and material suppliers/end-users to work closely together by means of a structured and scalable approach, enabled by a platform which connects stakeholders up to the execution of real scale trials – with the goal to implement solutions in real application scenarios and reduce market barriers; (B) enable different users (companies, public bodies, research organisations) to get easy access to advanced Pilot Lines and scientific expertise for know-how transfer enabled by the availability of these infrastructures; enable research institutions, SMEs and large companies to explore new application fields and start R&D work for markets that would otherwise not be addressed, by means of the funding programme to be launched in the second part of the project; (4) raise awareness and provide a solid marketing and Decision Support platform to extract, consolidate and broadcast the actual progress achieved in the different application streams to the community of material and product suppliers as well as research organisations in EU and beyond.

E. Public acceptance

Products carrying the BIOMAC brand will be defined as eco-friendly and will thus be widely accepted by the public. Currently, plastics undergo very slow degradation and may persist in the environment for decades. Thus, plastic products have been under scrutiny, due to associated environmental impacts and in particular marine pollution. [23] Plastics are currently associated with the drilling, transport, refinement, and use of petrochemicals, procedures with significant environmental impacts. Plastics produced within the BIOMAC ecosystem will circumvent the whole petrochemical value chain, using biomass as the raw material, which will further establish BIOMAC as an eco-friendly brand. The role in agriculture as well as the biodegradable in soil biomaterials for

[21] Jobs and growth generated by industrial biotechnology in Europe, EuropaBio 2016.

[22] http://www.reverdia.com/wp-content/uploads/Biosuccinium-PESP-Thermo-PU-Data-Sheet-R1003.14-EMAIL.pdf

[23] http://ec.europa.eu/environment/marine/good-environmental-status/descriptor-10/index_en.htm

agricultural applications are designed not only to be eco-friendly, but also to give added value to the soil, providing fertile carbon and ingredients that can improve the soil, e.g., biochar and antioxidants. Through a detailed dissemination and communication plan (outlined below) the public will be informed about the significant advantages of plastics produced with the BIOMAC technology, which will ensure wide public acceptance. Furthermore, customers concern about the nature of the materials, their functionality and applicability, performance, sustainability, and cost will be addressed through the dissemination plan. The BIOMAC project will work in the direction of public acceptance, focusing on activities that will lift safety concerns and demonstrate their functionality trough prototyping and other activities. By communicating distinctively to public that alternative feedstocks are used, which are not in competition with food, it is expected that polymers from BIOMAC will meet a higher acceptance than currently biopolymers made of food or feed-grade feedstocks.

2.1.2 Barriers that might affect the achievement of the expected impacts

The project partners have asked for EU funding being aware of the commercial risks of such an investment has and have tried to examine and mitigate the relevant risk factors adequately. Indicatively, possible barriers and obstacles are presented in Table 2.1. It is evident that although in the last two decades a high number of research projects have been dedicated to the development of new nano-enabled bio-based materials, however only a very limited number of such cases has been finally commercialized and reached the market. There are many reasons that these concepts have not been adopted by the market and this phenomenon is usually described as 'valley of death'.

Table 2.1: *Mitigation plans of BIOMAC for overcoming barriers for upscaling of materials concept*

Barrier	*How the project will help on its reduction*
Barrier 1: Lack of available infrastructures	
Pilot Lines are essential for upscaling materials concepts. Through them, laboratory results can be transferred to pre-industrial level by developing robust production technologies. However, such lines are not always available to companies since their acquisition cost is high, needs extra expertise in monitoring and control of lines, and SMEs are not eager to proceed in such investments unless they can decrease the risk. On the other hand, large industries, are not eager to stop their production lines to carry out tests for upscaling of new bio-based materials and products. BIOMAC ecosystem will provide a variety of flexible upscaling processes for testing, monitoring and process optimization of production of the new nano-reinforced biomaterials. This will support the rapid position of new materials in the market with a lower CAPEX due to the existing open access pilot lines that will be created through the BIOMAC Value Chain approach.	
Barrier 2: Industry acceptance – Investment interest	
A lot of highly promising innovative ideas often remain at Lab scale as they are not linked ether to standards and regulatory frameworks or remain disclosed ownership of RTDs. Especially, when it comes to materials containing NPs issues such as nanosafety are becoming a major barrier as the regulatory framework has not been yet assimilated by the industry. Indeed, end users, especially in the fields of packaging and automotive industry ignore products that are not harmonized with the standards. BIOMAC ecosystem will provide all the necessary services related to standardization, compliance with existing regulatory framework and nanosafety/standardization aspects of upscaling.	
Barrier 3: Industry needs characterization, standardized and validated solutions including Regulatory	
Characterization of materials and evaluation of the corresponding results is essential for the product development and upscaling. However, it is a common remark that this expertise exists mainly in universities and research centres. BIOMAC will provide technological, scientific and manufacturing expertise; through advance, accurate and fast characterization methods, combined with advanced modelling linking the crucial parameters of production steps with the final properties, lifetime and environmental impact of the new bio-based materials. A lot of highly promising innovative ideas often remain at Lab scale as they are not linked to standards and regulatory frameworks. REACH, SVHC list: Substances that may have serious and often irreversible effects on human health and the environment. OECD Safety of manufactured bio-based nanomaterials. The safety of Manufactured Nanomaterials is an important concern impacting regulatory bodies throughout the world. Due to their size, Manufactured Nanomaterials may require additional testing beyond the standard suite of tests used for other chemicals, to ensure that the impact on human health and the environment is fully understood. BIOMAC will give a special focus in the nanosafety, ISQ & UBU have experience in this field due to their participation in related actions (e.g., NanoReg). During the project, all necessary measurements for materials handling and use will be undertaken while the project will cluster with other corresponding initiatives (e.g., NanoSafety Cluster).	
Barrier 4: High perceived risk for investors	
Even the most promising ideas and concepts will probably need extra funding until they come to the market. Most often, SMEs cannot afford to invest on their own additional funds, while investors are reluctant to invest without securing the previous mentioned aspects (e.g., compliance with standards of the new materials, lifetime, performance). Inside the BIOMAC ecosystem proper conditions for bridging investors with SMEs will be provided. Thus, activities, such as translation of knowledge of discoveries developed usually in the academic community, in a context that is relevant to the industry investors will take place.	

Barrier 5: Public acceptance

Enterprises are aware of their customers' scepticism about adoption of new technologies and more specifically nano containing products. Main concerns of customers are on the nature of the materials, their functionality and applicability, their performance, their sustainability and naturally, on their cost. BIOMAC Ecosystem will work on the direction of public acceptance focusing on activities that they will lift nano-safety concerns, prove their functionality and appliance trough demonstration activities, certify their high performance and enhanced properties through standard testing results, explain the high sustainability achieved for both materials and products through LCA, contrast their high performance and multifunctionality to their initial production cost, etc. Planned dissemination and communication activities (see next paragraphs) will be focused on spreading awareness and explain the sector advances, in order to create a defensible position, penetrate this market, and gain competitive edge. As an example, the Construction industry is highly conservative, competitive, and price conscious with low profit margins.

2.2 Measures to maximise impact

BIOMAC project will address all issues related to exploitation, business development, innovation management, management of IP, data management as well as dissemination and communication activities. The dissemination and exploitation of the project's results as well as the management of knowledge and intellectual property are considered critical issues for the success of the project. A Dissemination and Exploitation plan, including a business strategy, plan, and model, has already been developed and will be constantly updated so that it encompasses all the progress that is made during the project. Finally, a communication strategy has already been projected with input from all partners and will be updated during the project. Further analysis is provided in the following subsections.

2.2.1 Plan for the Dissemination and Exploitation of the project results

A. Dissemination activities

The dissemination activities will begin with the initiation of the project already at month 1, all partners will have PMs which will ensure a high added value dissemination approach. The dissemination plan is designed to reach the largest audience possible, to inform EU stakeholders about the BIOMAC project's objectives and results as well as to build the basis of the sustainability phase post project. Dissemination activities aim at paving the way towards the future market penetration of BIOMAC solution and to ensure the successful exploitation of the project outputs (see 2.2.1). Moreover, they will promote the project's approach and encourage interested stakeholders and investors to adopt the solution developed and provide all serviced offered. The major aim is to ensure that the project's outcomes are widely disseminated to the appropriate target groups, at appropriate times (respecting the proposed project timeline), via appropriate methods (means of information disclosure, intensity, etc.). For a successful dissemination, all BIOMAC beneficiaries will cooperate in order to create higher added values at the market. In order to achieve this a precise **Ecosystem Dissemination Plan (EDP)** has been structured and will become the basis to achieve all actions set and to operate post project. With such an approach, BIOMAC PEDR (Plan for Exploitation & Dissemination of Results) will foster cross-border and cross-sectorial cooperation between the members of the consortium with the final goal to attract new customers to the BIOMAC ecosystem. All along with the project duration tactical, strategic, and operative elements of promotion, exploitation of results and awareness raising will be actively employed and will ensure capturing of a large audience, in an appropriate manner and hence ensure long-term sustainability of the project results and the established One Stop Shop. For an effective public outreach both standard dissemination activities, dedicated to public information, and more targeted engagement activities, where dialogue (communication) and alliances are created, will dynamically co-exist. The effectiveness of the proposed approach will be guaranteed also by the presence of important players representing BIOMAC value chain in the consortium that can take advantage of their network of contacts to raise awareness. The dissemination strategic steps are depictured in the figure below. Step by step the methodology will include (a) the definition of the subject to be disseminated, (b) the identification of the target audience that will be interested about project results and findings,

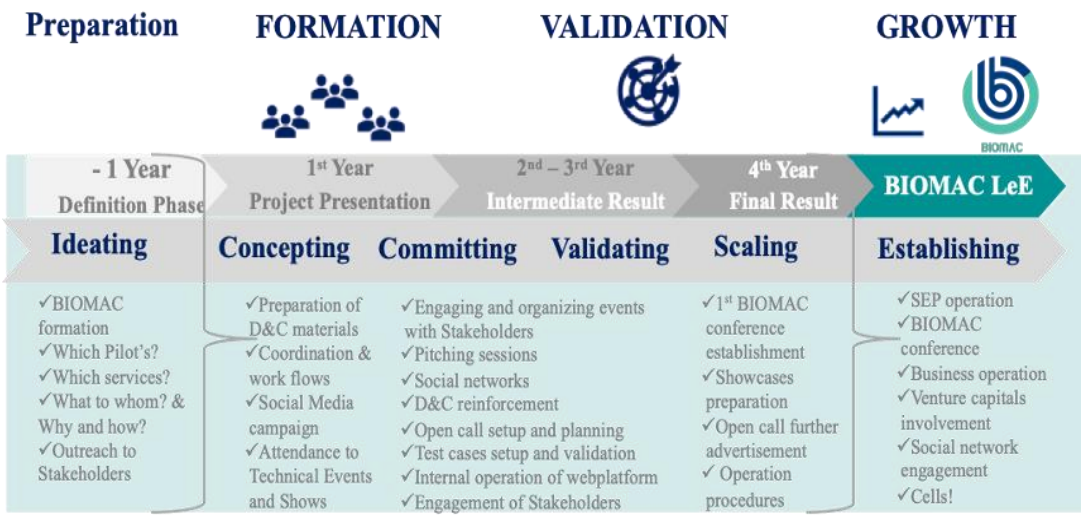


***Figure 2.2**:BIOMAC Dissemination Road Map.*

(c) the definition of the most effective ways (methods and tools) to reach the audience, (d) the appropriate timing to disseminate and (e) the dissemination policy and rules. In line with the EC Guidelines the dissemination activities will include all the best practices successfully tested by the partners in other projects: (1) The results and/or reports will be reviewed and sent to the relevant partners of the project before being published or disseminated. (2) On a two month basis, there will be an informative file to present to the partners the forthcoming events that are targeted to the particular partner. (3) All the consortium members will contribute to the project activities and will be informed about the final outcomes and the implication stemming from project results. All public results will be accessible from the project website and usable from all parties who may benefit Target Audiences: The BIOMAC consortium has identified ***11 groups of target audiences*** that would benefit out of the ecosystem:

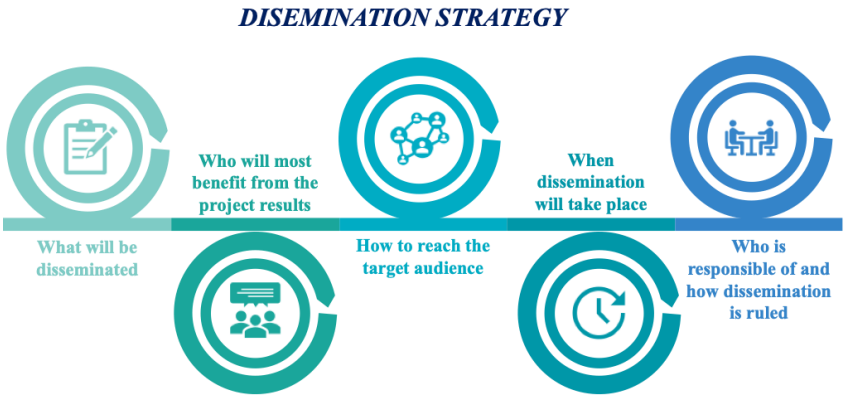


Figure 2.3: Ecosystem Dissemination Plan (EDP)

Table 2.2: Targeted audience groups.

Category	Target group
European Start-ups	SME, and Midcap communities: Companies (tech and non-tech) interested in understanding the potential of nanoreinforced biomaterials
Industry	Providers of materials and process solutions (e.g., chemical companies, equipment), Companies involved End users in the fields of automotive, food packaging and other associated sectors where technologies can be applied, High technology SMEs interested in adopting new technologies/knowledge and starting new business models.
Regional innovation networks/accelerators	Existing innovation hubs interested in expanding existing local support to European cross-border support. Ecosystems that operate and are related to polymers and PE.
Design stakeholders	Owners of products and services, platform providers, inventors, R&D entities
Cross border community	Participants in other projects in the call. Policy makers: Entities in charge of funding programmes.
Broader innovation community	Interested in the best practices for the advance of the EU sector. High technology SMEs interested in upscaling new materials and starting new business models.
Investor	VC, Angels, Crowd funding platforms, Bank looking for innovative companies interested in embedding new innovative components in their products and services.
End users	in the fields of automotive, construction, food packaging and printed electronics sector and other associated sectors where technologies can be applied, replicating results
General public	Individuals including youth
R&D communities	Universities and research centres, participants in related EU projects, research societies, interested in the bio-based materials sector
Regulatory authorities	Ministries, organisations, associations, regulatory authorities, NGOs, non-profit organizations, public initiatives, Standardization and certification bodies

Dissemination channels: General and specific dissemination channels will be utilized throughout the project's execution, that also includes the respective schedule for the development of these activities. The EC financial support to the project will always be highlighted in the dissemination activities, and the logo of the EC will be included in all activities. All relevant online and offline dissemination activities shall be reported to the Dissemination WP leader (EUBIA), in order to ensure an integrated and cohesive communication. This will apply especially in the case of partners attending fairs, exhibitions, conferences or other public events. A **draft Dissemination Roadmap** has already been developed (Table 2.3), that will lay the ground for the project's dissemination plan which will be updated on a yearly basis. Table 2.3 also shows a set of adequate indicators (Dissemination KPIs) to measure the impact of dissemination activities.

Table 2.3: BIOMAC Dissemination & Communication Roadmap and relevant KPIs (Conferences and targeted journals List)

Dissemination activity	Target audience*	Expected Impact	Measure/Target (KPIs)
Consortium activities			
Project Website	GP HT, EU, RNA, BIC, RC	Awareness of the BIOMAC project and its scope	- 50 visits per day after year 3 - Receive 5 emails via website per week

Printed material	HT, TC, EU, BIC, GP, RC	Attract interest of industry, including SME in the Ecosystem	- Development of 10 printed concepts - Distribution of 5.000 leaflets in events - Express of interest from 250 stakeholders
Presence in social media (SM)	HT, TC, EU, BIC,	Attract interest of industry and connect with stakeholder pages and accounts in SM	- 1000 connections in LinkedIn account by the end of the project - 5 posts per month in SM accounts
Videos posted in website, YouTube	GP, IN, AN, HT, TC, EU, RC	Attract interest of industry (mainly SMEs) and investors by showcasing BIOMAC	- 2 videos presenting the PLs - 2 videos presenting the services - 5 videos of success test cases
Press Releases	GP, RC, IN, AN, RC, PM	Awareness of the BIOMAC Ecosystem and capacities of it	- 2 issues per year in printed publications - 2 releases in TV in 3 different countries
Participation in events	RC, BIC, IN, AN, HT, TC, EU, RNA, PM	Liaison with stakeholders and present the BIOMAC Ecosystem	- Participation if 30 events (fairs, conferences, etc) during the project, having booths is at least 5 of them
Legal Entity activities			
Ecosystem Web platform	HT, TC, EU, RNA, IN, AN, PM, BIC, RC	Presentation of Ecosystem Services and terms, acceptance of requests as SEP	- 150 visits per day by the end of project - 10 request during the 4th year - Presentation of 5 success stories
BIOMAC conference	RNA, BIC, IN, AN, HT, TC, EU, RC, PM	Establishment of a biannual conference in the field of bio-based nanomaterials	- 1 international conference at year 5 - 10 invited speakers, 60 presentations - 300 attendees in the 1st conference
Workshops/ training/Stakeholder events	HT, TC, RC, BIC	Awareness of BIOMAC technologies and success stories	- 3 workshops in international events - 50 attendees/workshop - 3 Stakeholder meetings showcasing the successful Test Cases (GR, IT, DE)
Online Journal	HT, RNA, IN, AN, PM RC	Presentation of new technologies, success stories to stakeholders	- 6 issues per year after Year 4 of the project

Conferences and other related events to BIOMAC	
Conferences on technology development	Congress on Materials Science & Engineering, ENERGY in BUILDINGS & TRANSPORTATION, Biopolymers, The Plastics Show, Manufacturing & technology tradeshow, EuroNanoForum and Industrial Technologies. (e.g., ISGC series, La Rochelle, France), JEC Paris, MRS, ACS Green Chemistry Conference, Gordon Research Conference. Polymer processing Society, BioEnviromental Polymer Society, BioFor & PaperWeek Canada. European Bioplastic Conference, AMI conference
Conferences on innovation management	Innovation Roundtable® Summit, Academy for Design Innovation Management Conference, Innovation, Management and Technology Conference, Innovation Events and Open Innovation Conferences, R&D Global Innovation Forum, Angel Investment Day – HunBAN.
Trade fairs	Global Automotive Materials Congress, Intelligent Traffic and Transportation (ICITT), Industrial Technologies, HANNOVER MESSE, European Trade Fair and Forum for Composites, JEC WORLD the Leading International Composite Show, BIOFACH, K- Fair,

Journals related with BIOMAC			
Journal	**Imp. Fact.**	**Journal**	**Imp. Fact.**
Journals on (Nano)Materials and Biomass Valorisation			
ACS Nano	13.709	Green Chemistry (RSC)	9.405
Journal of Nanomaterials (OA)	2.504	ChemSusChem (Wiley)	7.804
Bioresource Technology (OA)	5.807	ACS Sustainable Chemistry & Engineering	6.97
- Journal of Cleaner Production	6.395	Enviromental Science and Technology	7.149
Business Management related Journals			
Creativity & Innovation Management	2.015	Journal Of Entrepreneurship Innovation	3.905
Journal Of Innovation Management	2.305	European Journal Of Innovation Management	1.385
Journal Of Product IM	4.305	Innovation-Management Policy & Practice	0.955

*HT: High-tech companies, TC: Tech companies, EU: End Users, RNA: Regional networks and accelerators, IN: Investors, VCs, AN: Angels, PM: Policy makers, BIC: Broader innovation community, GP: General Public, RC: Research Community

Knowledge Management and IPR Protection

Consortium Agreement: A Consortium Agreement (CA) will be agreed and signed prior to project commencement in order to define the important points necessary to obtain the best possible management (financial conditions, IPR, planning). The DESCA H2020 template will be used as reference. Prior to the CA definition, the main guidelines for IPR management will be defined. Being one of the keys for the project success, background knowledge of each of the partners that may be used to achieve the project objectives will be identified and measures will be taken in order to deal with the use of Intellectual Property. Moreover, the consortium agreement will include any key steps that need to be taken for the best management and exploitation of the project. The IPR manager will be responsible for the management and exploitation of Intellectual Property Rights (IPR). The DEC Committee (IEM, IPRM, DCM)

will evaluate actions related to technology transfer under the innovation management plan, promoting all technologies validated to existing customers/clients of the consortium members and also identifying additional related stakeholders. After a recent search, it is the consortium's opinion that the proposed technology does not contravene any existing patents (see section 1.4). During the project meetings the internal results will be reviewed with the goal of identifying important ideas and defining an individual strategy for the positioning of these ideas in the standardization, training and commercialization processes.

Management of knowledge and foreground IPR: The consortium agreement will ensure the protection of the Intellectual Property of all partners. The IPR arising from the results of the project will be the property of the partner that has developed it. However, knowledge needed for the completion of the project will be made accessible among partners. During the project, scientific articles and case studies that are produced will be published whenever sensitive information is not included upon agreements of clients and the ecosystem. Due to the different policies of each member of the consortium, Gold and Green model will be considered for open access to maximise public and researcher access. Consequently, open access will be granted to all scientific publications resulting from the BIOMAC project however, protected by copyright statements. On the other hand, procedures and tools that will be produced will remain on their respective owners for exploitation. If the results of the project are eligible for patenting, actions will be taken at the earliest possible date. In cases of joint inventions, the ownership will be shared in accordance with a notified agreement.

Background IPR: The Background knowledge of the partners, as stated before, will be identified prior to the writing of the Consortium Agreement. Background IPR will remain the property of the Originator Company or organisation. However, where needed, during the period of the project, all "background" knowledge of the partners necessary for carrying out the project going to be available for use by any other partner of the consortium. Access rights will be granted, royalty-free, by the owning partner. In terms of post-project exploitation, access will be once again granted by the owning partner, either royalty-free or by charging a license fee under reasonable terms, promoting in that way the collaboration among nations and the advancements of the research and innovation. For better protection of the knowledge shared, a Non-Disclosure Agreement could be signed between the involving partners. Finally, all knowledge will be managed in accordance with the Grant Agreement and a Consortium Agreement that will be timely prepared and signed by all consortium members.

Patents and Knowledge Sharing: If considered appropriate, the "results" IP on new prototypes developed will be patented in Europe and in any relevant region of the world. Rules for governing access rights for the jointly executed research: Partners working together for achieving specific objectives of the project will enjoy access rights to: (A) The knowledge arising from work carried out under the joint activities and (B) The pre-existing know-how of the other participants when it is necessary for the project implementation.

Data management plan (DMP)

The partners have considered issues related to management of research data as they want to ensure data security, accessibility, and validation of results. A Data Management Plan (DMP) will be conducted as part of a concrete research data management strategy which will minimize possible duplication of effort and the risk of data loss and will offer integrity to the data (month 06 and updated by month 48). Furthermore, the research efficiency will increase, time and resources will be saved in long term and compliance with practices conducted in industry and commerce will be achieved. The DMP will describe plans for creating, organising, documenting, storing, and sharing data. It will take into account issues such as data protection and confidentiality, data preservation and curation and will provide a framework that will support researchers and their data throughout the course of their research and beyond. It will be constantly updated, describing what kind of research data will be generated, what policies apply to the data. Funding and legal policies and data management practices as backups, storage, access control, archiving will be determined. It will be clear who will own and have access to what data and who will be responsible for each aspect of the plan. Moreover, partners will try to clearly describe which data will retain value after the end of the project and how its reuse will be enabled and how the long-term preservation ensured after the original research is completed. Partners will share research data generated allowing others to replicate, validate, or correct their results, thereby improving the scientific record. This will increase the research's integrity and replication as those who make use of their data and cite it in their own research will disseminate the results possibly in other disciplines, sectors, and countries. Moreover, in the future, partners will be able to identify, retrieve, and understand the data themselves after they have lost familiarity with it, perhaps several

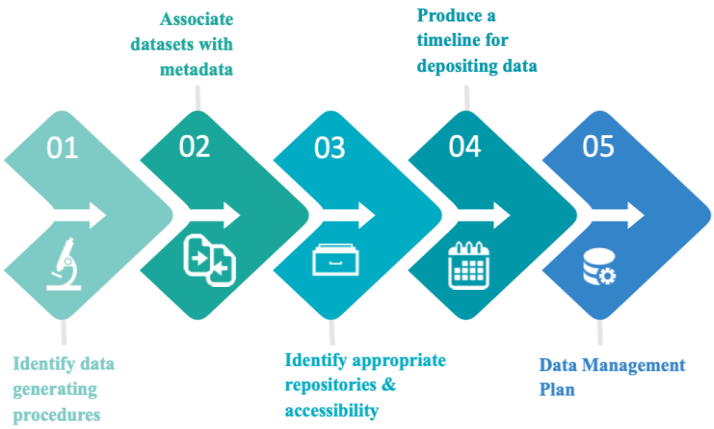


Figure 2.4: *Data management plan strategy of the project*

years hence. Partners will not share specific parts of their research data if the achievement of the project's main objectives would be jeopardized by making those specific parts of the research data openly accessible. They will not also share data that they may not have the legal right to share, maybe because the parts of them belong to other authors or entities. The methodology and standards, whether and how these data will be shared and/or made open, and how it will be curated and preserved will be decided by the consortium during project implementation. The DMP will follow the template demonstrated below for each of the datasets generated:

- ✓ Data set reference and name: identifier for the produced data.
- ✓ Data set description: description, nature, scale, information about integration and reuse etc.
- ✓ Standards and metadata: reference to the standards that will be used and how the metadata will be created.
- ✓ Data sharing: access procedures, software or other tools needed for access, repository, reasons for confidential data Archiving and preservation: procedures for the long-term preservation of the data, estimated end volume.

B. Exploitation activities (*Exploitation plan for the BIOMAC Ecosystem & End Users)*

The exploitation activities of the project concerns two main directions. On one hand there is the exploitation of the **main result** of the project that is the BIOMAC OITB itself that will be elaborated at WP8, and on the other hand the exploitation of the end users that will be elaborated in WP5. The consortium has worked on the exploitation plans for both cases. Initially the Business Plan of OITB is presented in B1-B5 and then a brief description of the exploitation routes of the end users follows.

B1. Value proposition and role of the *BIOMAC* Ecosystem

The value proposition of BIOMAC OITB is the **operation of a new, open innovation ecosystem targeting in the upscaling of nano-enabled bio-based materials**. This will be achieved via the collaboration of high expertise groups across Europe in the field of biomass pre-treatment, value platform chemicals extraction, processing and manufacturing, monitoring, testing, characterization, modelling, environmental assessment, standardization, nanosafety, innovation management and training. ***BIOMAC* Ecosystem will provide its services to SMEs, midcaps, large industries and R&D entities** interested in developing new applications and products based on **nano-reinforced bio-based polymers.** Besides this main function, the BIOMAC OITB will offer additional business opportunities to its clients through the developed open innovation environment. On one hand it will act as a **Pan-European association (BIOMAC Community)**, serving the interests of its members regarding the field of nano-enabled bio-based materials and products offering opportunities of networking, technology transfer, reduced fees to training workshops and events etc. Finally, **the OITB will serve as a facilitator of IPR and technology transfer services**, focusing on the technologies that will be developed inside the Ecosystem. All three activities are expected to contribute to the revenues of the OITB as a whole.

B1.1) SWOT & PEST analysis:

The consortium has discussed about the venture of the BIOMAC ecosystem and analysed internal and external factors that can influence the BIOMAC ecosystem as well as the external macro environment in which the ecosystem will be positioned. The following table presents the SWOT and PEST analysis. Based on this, the concept of the BIOMAC ecosystem is aligned positively with forces of change affecting its positioning environment.

Table 2.4: *SWOT and PEST analysis of the BIOMAC concept*

The SWOT-PEST analysis			
Strengths	**Weaknesses**	**Political**	**Economic**
- Single Entry Point (SEP) - Cross sectoral market approach Value Chain - Unique combination of EU SoA PLs - Reduce time, cost risk to market through the use of BIOMAC ecosystem - New smart biomaterials value chains - Cross sectoral market approach (biomass, chemicals, packaging) - New bio-based value chains	- New concept with uncertain customer acceptance - Necessity of strong IPR policy within the ecosystem - Maintain collaboration between the ecosystem key partners - Establishment of partner agreements and liability obstacles	- European strategic goals for global competition in KETs [24] - EU policy and legislation for nanomaterials [25] - Bio Based Industries (BBI) roadmaps - European strategy for plastics [26] - EU policy for circular economy - EU policy for reducing agricultural waste burning	- Financial support for clients to use BIOMAC services - Cost of raw materials/biomass - Costs involved in the life cycle - Financial support for clients to use ***BIOMAC*** services - Seed funding for product commercialization

[24] https://ec.europa.eu/growth/industry/policy/key-enabling-technologies/european-strategy_en

[25] https://publications.europa.eu/en/publication-detail/-/publication/be32dfc7-1499-4328-b54f-a9f024805f59/language-en

[26] http://ec.europa.eu/environment/waste/plastic_waste.htm

Opportunities	Threats	Social	Technological
-High growth industrial sectors selected -Develop products and supply chains to capture immediate market needs - Enhance productivity and efficiency of bio-based solutions making them more affordable in mass production - Cooperation with existing companies offering niche services - Extremely high market potential	- Threat of new entrance of competitors in the EU market - Fuzzy compliance with regulations on bio-based nanomaterials - Customer reaction to pricing policies - End users are relatively reluctant to accept new technologies - Open Innovation - Need of strong IPR policy - Target markets to re-learn "value selling"	- Public awareness on economic and environmental benefits of the BIOMAC technologies - Reduction of greenhouse gas emissions by using no fossil-based materials - Reduction of farmer exposure to harmful fumes from burning agricultural waste - Transition towards circular economy	- Better use of raw materials and resources with reduced environmental impact - Upscaling of manufacturing lines of nanomaterials and bio-based polymers - Knowledge transfer from pilot lines to industrial processes - Materials and processes logistics - Waste treatment technologies

B1.2) Canvas Business Model:

An initial approach based on the Business Model Canvas, operating as a basis for the business plan full development, is presented below trying to demonstrate the factors that explain how the BIOMAC project results will successfully penetrate the targeted markets.

Table 2.5: *BIOMAC Canvas Business Model*

Key partners	*Key activities*	*Value proposition*	*Customer relationship*	*Customer segments*
- Other OTIBs - EU Pilots - Investors - Venture Capitals - EEN - Biomass providers - Data repositories - European Clusters and Associations - EPPN, Nanosafety, EMMC Public stakeholders (EU, governments, Industrial Associations etc.) - Banks/Financial Institutions	- IBB acting as SEP - Technology scale up, upgrade and validation - Commercialization support - Monitoring and modelling - Clustering activities - Design and development of PILOT technology - Value chain management ***Key resources*** - Experience in process engineering - Existing Pilot Lines - IPR - Infrastructures	✓ **Open Access to 17 Pilot Lines** ✓ **Offer and commercialization of transversal Support services** ✓ **Upgrades and monitoring of PLs** ✓ **Training activities** ✓ **Feasibility studies** ✓ **Prototypes** ✓ **Brokerage events** ✓ **Funding landscaping** ✓ **Open Innovation Community**	- Single entry point SEP - Co-creation - Dealers agreements ***Channels*** - Web platform - Collaboration with other OITBs	<u>Primary:</u> Automotive Construction Agriculture Food packaging Printed Electronics <u>Secondary</u> Plastics manufacturers (packaging, piping, etc.) Materials manufacturers Chemical industries
Cost structure		***Revenue Streams***		
- Costs for maintenance and upgrading of Pilot facilities and infrastructures - Consumables and fixed costs (materials, salaries) - Cost for experienced staff - Marketing costs - Large scope of operation		- Fees of access to PLs and services - Community membership fees - Innovation consulting - Conference and pitching event fees - IPR protection-patents-licensing of processes to companies		

B2. Management and organization of Ecosystem

The members of the BIOMAC ecosystem have already decided that the most efficient way for the proper representation and operation of the Ecosystem is to use IBB that will undertake the whole administrative, financial and legal issues of the SEP with the assistance of the OITB members (AUTH, EUBIA, MISC, LTU, BBEPP, LIST, RISE, ATB, UEdin, AIMP, FH-WKI, CNano, ITEN, AIMEN, PoliMi, DTI, NNT, IDEN, AXIA, UBU, Abis, EXEL, EUBP, UPD, IRIS, RDC, ISQ & STAM). The management and organizational structure of the OITB and the SEP has been described in sections 1.4.1 and 2.0.

B3. Legal & Operational Plan

The major aspects governing the operation of IBB as a SEP and all individual relationships with the members of the BIOMAC ecosystem, as well as the clients, are presented below and will be finalized in WP1, WP5 and WP8.

Access to the BIOMAC Pilot/services by the end users will be based on an agreement which the IBB will conclude with each individual customer. The **End Users Agreement**:

✓ The SEP provides access to Pilots and services to properly registered users via an online platform within the scope of a technology services solution.

✓ The services are provided to the User against payment of a fee, agreed with the actual service provider members of the BIOMAC Ecosystem, for the duration of its use ("pay-per-use" business model).

✓ The SEP reserves the right to improve, adapt or modify the online platform in terms of content and/or structure. For example, the SEP may modify or discontinue some services, make available new ones, and modify or extend the user interface of the online platform. In doing so, the SEP will consider the User's legitimate interests.

✓ The SEP will also offer a customer support service thought the web platform. Also, options for "after sales" services will be projected on the End-User Agreement.

The SEP shall ensure that all end users intending to have access to the Pilot/services accept the End-user Agreement. Concerning the relationships of the SEP and the core members of the OITB, these will be defined by **the Framework Agreement.** The **Framework Agreement (FA) will be established between the core members of the OITB and the SEP** and will form the contractual framework for the provision of the BIOMAC services via the SEP. In the FA, the services that each core member will provide will be described in detail (e.g., access to Pilot Lines, modelling, characterization, consultancy etc) as well as other general terms (e.g., legal aspects, confidentiality terms, organization of work, pricing policy etc). Each Party of the FA may – with the mutual agreement of the respective other party – amend or add information to the pilot/services covered by the FA. New pilot/services will be welcome, provided that the terms and conditions set forth in each new Pilot/service sheet do not contradict those of the Agreement. The FA will include a **Statement of Work (SoW)** that will be activated each time a new customer will require the services of the OITB. In the SoW that will be signed between the SEP and separately by each involved member of the OITB, the details of the required service for the specific end user will be provided (analytical tasks, costs, persons involved, resources required).

B4. Market segmentation, size & needs

BIOMAC has conducted already a **Discovery market search** per End User Sub market where the CAGR, the market size and value has been defined (See also 1.4). [27]

Table 2.6: *BIOMAC Targeted Market size and value*

Market	Sub markets	*CAGR*	*Market Sub markets Market size and value*
Automotive	Interior furniture component	7.0%	The global polyurethane market size is expected to reach 105.2bn by 2025, according to the report by Grand View Research. Rising awareness regarding resource conservation and optimum manufacturing processes and low labour costs in developing countries are factors estimated to fuel overall market demand through 2025. Growing demand for bio-based polyurethane (PU) as well as demand for innovation and sustainability are key factors driving the industry. Rising government interferences in order to reduce greenhouse gas (GHG) emissions are also estimated to have a positive impact on overall market growth. [28]
Food Packaging	Food packaging flexible films	20,9%	PLA has the second largest production volume of any bioplastic. PLA represents the alternative to cost-efficient, non-petroleum plastic. A significant benefit of PLA as a bioplastic is its versatility and the fact that it naturally degrades when exposed to the environment. [29] The global polylactic acid is projected to reach 5.16 Billion EUR by 2020, growing at a CAGR of 20.9% during the forecast period 2015-2020. The last 5 years the PLA Market size was above 500 kilo tons and is poised to grow at significant growth rate. Packaging segment accounted for the maximum volume share 16 owing to rising adoption of green packaging among consumers.
Construction	Modular pedestrian bridge Civil Engineering	10.6%	The global construction market reached a value of nearly 12bn in 2018, having grown at a compound annual growth rate (CAGR) of 8.3% since 2014, and is expected to grow at a CAGR of 10.6% to nearly 18 bn by 2022. Growth in the historic period resulted from strong economic growth in emerging markets, low interest rates, and technological advances. Factors that negatively affected growth in the historic period were geo-political tensions that hampered construction activity globally. Civil engineering construction market is segmented into utility system construction and bridge construction and civil engineering construction.

[27] https://discovery.patsnap.com

[28] https://www.grandviewresearch.com/press-release/global-polyurethane-pu-market

[29] https://www.creativemechanisms.com/blog/learn-about-polylactic-acid-pla-prototypes

Agriculture	Agricultural films	7%	The global market size of plastics for agricultural use is 5.6 MT, with an average annual growth of 7 % with peaks of growth in Asian areas where the demand of increase in food production is stronger. [30] Total agriculture plastics market in Europe is 674 ktons, 46 % used for vegetable productions. [31] Mulch films amounts for 86 kton/year, only the 3,5% is biodegradable and the 9,4% oxo-degradable. Mulch films are mainly used in Italy, France, Germany, Benelux and Spain.] Among biodegradable biopolymers PBAT and starch blends are as the most relevant.
Printed Electronics	Flexible wearable sensors	18.8%	Wearable sensors are expected to gain traction over the forecast period. The application of these sensors in fitness bands is estimated to hold largest market share in 2016. Global Flexible Electronics market accounted for $5.53 billion in 2017 and is expected to reach 24.78b by 2026 growing at a CAGR of 18.1%. Wearable sensors are a vital aspect of wearable devices owing to growing focus on health and fitness monitoring activities. Wearable sensors are emerging as a revolution in monitoring healthcare and fitness and reduce hospital cost for patients. [32]

B5. Sustainability of the Ecosystem & Financial projections and KERs

The 28 core members of the BIOMAC Ecosystem will provide their services to customers in the field for many years. The partners have agreed that it will be beneficial for them to join forces through the formation of an Ecosystem as on one hand they will share administrative costs (will be undertaken by IBB), and on the other hand they will develop synergies that will result in better and faster provision of services to their clients. Moreover, the members of the Ecosystem will shift from a regional activity to a Pan-European market through the SEP. This is expected to have a significant increase of their revenues. Additionally, based on their current experience, BIOMAC partners have also identified the three most crucial risks for the successful accomplishment of the Ecosystem's creation.

B5.1) Main Risks to sustainability and contingency plans:

A) Access to finance: The BIOMAC project aims at proposing an integrated access to a large diversified ecosystem of actors through a SEP. Considering the potential barriers in the access to finance in order to guarantee the sustainability of the outcomes of the project – specifically the SEP and related services carried out through partners – we mention the potential risks for funding providers connected to the positioning of the offering of the BIOMAC solutions in the competitive scenario and the commercial features of the offering. The value proposition of the offering could be underestimated by funders or their perception of the risks associated with the new technological processes proposed – within the open innovation paradigm – could be particularly high. These barriers could be overcome with an appropriate and detailed business plan, also based on interaction and feedback obtained throughout the project (WP5 and WP8). Another barrier could be related to the scale of the financial resources, that is access to a more limited amount of resources than the necessary one, or in the timeframe requested. To overcome this issue, the project could foresee communication activities and promotion of the OITB and its services at wider international scale towards appropriate funding actors.

B) Pricing Policy: The definition of the pricing policy is specifically one of the activities and related deliverable of the project. One barrier in fully defining an appropriate pricing policy is connected to the need of balancing accessibility to the services offered through the SEP towards the OITB (and SEP), and the economic sustainability of the ecosystem. More precisely, the barrier is related to the organizational effort (and coordination activity) required to overcome any bargaining power of actors within the ecosystems, to aligning price definition with value creation and distribution among partners. This is connected to the barrier related to the variety of actors involved in the ecosystem with different cost structures that should be considered in the SEP. From this point of view, specific attention will be given within the project (WP1, WP5, WP6, and WP8) to promote information exchange among producers in relation to cost structures. Another barrier is related to the need of developing a differentiation pricing policy based on customers (large vs. SMEs) and link to the IPR regime that the project will outline. This issue will be overcome by exploiting the ***open call within WP7*** to fine-tune the scheme of the pricing policy through interaction with potential customers.

C) Internal Conflicts: Internal conflicts may rise in relation to a shared IPR strategy able to guarantee innovation but also appropriability of the results of innovation and related production activities. Within the project, in the development of the SEP, IPR definition will be crucial to avoid problems *ex-post* where unclear IPR policy could generate conflicts within the ecosystem on knowledge protection or economic appropriability of the results developed through the OITB. The possibility to articulate IPR strategy according to multiple commercial and technological scenarios by involving directly the partners should reduce such barrier to sustainability. Internal conflicts could also be related to the access to market through the SEP and eventually the priority among partners within the ecosystems

[30] Reynolds A., Trends in the agriculture film business, AMI Agriculture Film 2018, Oral presentation, 2018

[31] Plasticulture, N°135, 2016, B. Figuier

[32] The "Flexible Electronics - Global Market Outlook (2017-2026)" report has been added to ResearchAndMarkets.com's offering.

in relation to promising commercial opportunities (i.e., large demand). From this point of view, activities related to different WPs (1, 5, 6 and 8) should be directly oriented to share information and promote discussion and collaboration among the partners on their perspectives and expectations concerning market opportunities, in order to validate a clear, agreed and approved process of distribution of product development across pilot lines. This will help reducing competitive behaviours internally.

B5.2) Financial projections:

BIOMAC targets clients across the supply chain, starting from **producers of biomass, chemical companies processing biomass or derivatives and end users** (e.g., food packaging, automotive, construction, agriculture, or printed electronics) as well as R&D entities eager to further investigate innovative concepts at a larger scale. It should be highlighted that as the global bio-based materials market is going to witness a heavy demand the next decade due to growing concerns regarding environment, the services of BIOMAC will be extremely attractive to the industry. According to recent market reports, the global market for bio-based materials was estimated at 9.9b € in 2017 and is expected to reach 85b € in 2026, growing at a CAGR of 27.1%. Main consumers are the markets of rigid & flexible packaging, automotive & transport, consumer goods, building & construction, electrics & electronics and textiles. [33]. Based on this fact, BIOMAC is positioned in a fast-growing sector with increased R&D expenditures [34]. Already during the proposal preparation, a significant number of letters of intent (LoI) from potential clients have been collected (a total of 30), proving the interest of the industry in what BIOMAC offers. The Innovation Management team of BIOMAC, in collaboration with the other partners, have worked on a draft estimation of the financial projections that the BIOMAC project will have during the 5 first years of its sustainability phase. The presented costs are the real costs provided by the core partners for the provision of similar services. The revenue model of the OITB includes the following sources:

A. Fees for accessing the PLs and services. The average cost for a test case will be ~200.000€. This cost concerns the access to PLs and the supporting services for a duration of 6 months. The cost breakdown (in percentages) in the various services of the OITB for a test case is:

Percentage of cost	*Cost*
45%	for the use of the Pilot Line, adjustment of monitoring systems and raw materials
32%	for the evaluation of the materials and products (characterization, testing and field tests)
10%	for conduction of simulation and materials/process modelling
5%	for the H&S issues (including nano-safety) as well as environmental assessment
8%	for the industrial implementation issues (e.g., business planning, marketing etc).

These percentages may vary depending on the client's preference on the services of the OITB (for example a client might have already proceeded in-house to an environmental assessment or a preliminary business plan and thus will not be interested in the relevant services).

B. Fees for the BIOMAC community participation. The members of the community will pay an annual fee in order to enjoy the benefits that the community will offer. The exact amount of fee will be determined during the project, but it is expected to be in the range of 3.000€ per year per partner.

C. IPR & technology transfer facilitator. The open innovation environment that will be created around the BIOMAC OITB will provide a prosperous ground for collaboration between the clients. The OITB will provide consultancy services facilitating such interactions and agreements, charging a fee to each success case. It is expected that the OITB will have a profit of ~150.000€ per contract for these services.

Based on the interest of stakeholders already expressed through the LoI, as well as the targeted dissemination and marketing activities that will be undertaken during the project's duration, the partners estimate that for the first year of the operation of the Ecosystem, **close to 25 test cases will be introduced** and there will be an increase of interest for the next year, reaching more than **100 test cases by the fifth year**. The table below shows the preliminary financial projections for the members of the Ecosystem. A 2.5% inflation rate per year has been considered for the calculations. Based on Table 2.7, it is expected that by the fifth year after the project ends, the BIOMAC OITB will have a cumulative revenue of **76.093.462€, that is 5 times the EC funding for the project**, reaching thus the targets set by the work program. Taking into account a new job for every 200.000€ of additional revenue, it is expected that **~380 new jobs will be created within the 28 Core partners of the Ecosystem**. However, the impact on jobs creation will be much higher when also including the new jobs positions concerning the **clients** that will enter the ecosystem

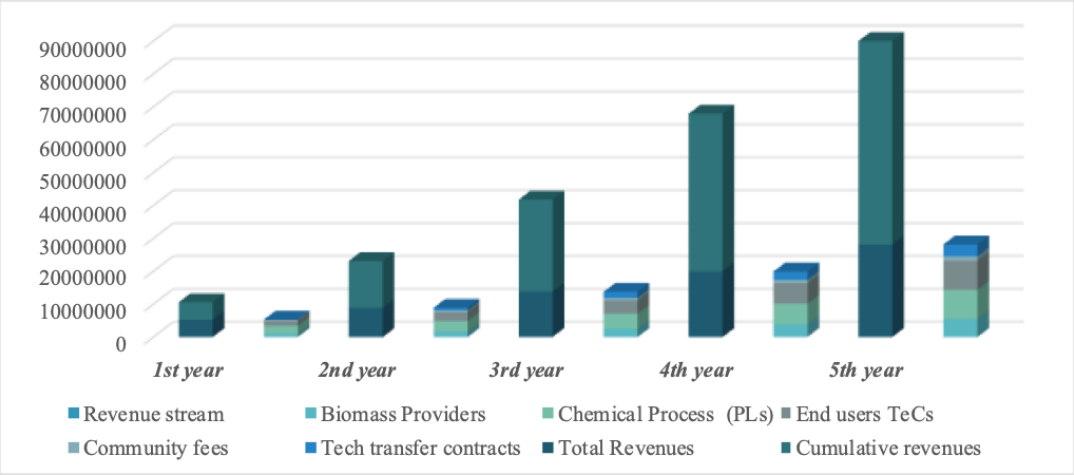
[33] "Global bio-based materials market" Zion Market Research, August 2018

[34] D. VIaggi, "The bioeconomy, delivery sustainable green growth", CAB International, 2018

during these 5 years period, that is estimated to reach **300**. With a moderate estimation of creation of ~4 new jobs for each of the clients entering the Ecosystem (due to their business growth attributed in novel and competitive products they will develop), a total of ~**1200 new jobs** will be created.

Table 2.7: BIOMAC sustainability and business plan of Core partners

	1st year		2nd year		3rd year		4th year		5th year	
Revenue stream	**# contracts**	**Revenue**	**# contracts**	**Revenue**	**# contracts**	**Revenue**	**# contracts**	**Revenue**	**# contracts**	**Revenue**
Biomass Providers	5	1.000.000 €	8	1.640.000 €	12	2.521.500 €	18	3.876.806 €	25	5.519.064 €
Chemical Process (PLs)	12	2.400.000 €	16	3.280.000 €	22	4.622.750 €	29	6.245.966 €	40	8.830.503 €
End users TeCs	8	1.600.000 €	13	2.665.000 €	19	3.992.375 €	29	6.245.966 €	40	8.830.503 €
Community fees	100	300.000 €	180	553.500 €	240	756.450 €	310	1.001.508 €	400	1.324.575 €
Tech transfer contracts	0	- €	5	768.750 €	12	1.891.125 €	16	2.584.538 €	22	3.642.583 €
Total Revenues		5.300.000 €		8.907.250 €		13.784.200 €		19.954.783 €		28.147.229 €
Cumulative revenues		5.300.000 €		14.207.250 €		27.991.450 €		47.946.233 €		76.093.462 €



Individual Exploitation plans for Core partners and End Users of the Ecosystem

Besides the business plan for IBB (BIOMAC SEP ecosystem), the partners have also worked on their preliminary individual exploitation plans. All partners of the BIOMAC project expect to have significant advantages by exploiting the results of the project. The following table briefly presents the individual exploitation routes for the BIOMAC partners. For simplicity reasons we have proceeded in groupings of partners with common or similar exploitation routes. In Table 2.8, the individual exploitation routes for the core members of the ecosystem are presented.

Table 2.8: Individual exploitation routes of core members of Ecosystem

Exploitable results and new knowledge/expertise	*Partner (s)*	*Exploitation plan/route*
- Upgraded pilot lines, capable to produce nano-enabled bio-based materials - Methods for production of nanomaterials from biomass	PL owners	Open access to PL, rent of equipment for upscaling and testing via the SEP agreement Licensing fees Further upgrade of PLs
Methodologies for developing protocols for nano-enabled bio-based materials	PL owners	Services for characterization of nano enabled surfaces and products via the SEP agreement
Materials Characterization protocols validated in real scale scenarios	PoliMi	Services for characterization of bio-based materials and products via the SEP agreement
Knowledge and Expertise in applying monitoring and process control systems in Pilot Lines	PL owners	Installation of monitoring systems in lines via the SEP agreement, Licensing of technologies
Expertise in application of monitoring methods for materials	IRIS, Abis, ISQ, STAM	Provision of services to clients of the Ecosystem via the SEP agreement
New simulation models, MODA workflows and ROMs for PL processes including NPs	IDE, UBU, RDC, ISQ	Provision of services to clients of the Ecosystem via the SEP agreement. Licensing of developed software
Knowledge in monitoring and risk assessment of H&S of PL, including nanosafety	UBU, UEdin	Assessment of H&S risk assessment to clients of pilot lines via the SEP agreement
Data Management and Platform for storage and acquisition	IDE, RDC	Provision of services to clients of the Ecosystem via the SEP platform
Knowledge on LCA/LCC and Nanosafety	ISQ/UBU	Service provision via the SEP agreement
Development of Value chain Decision Support tool	AXIA, UNIPD	Provision of services to clients of the Ecosystem via the SEP platform

Knowledge on standardization and regulatory of bio-based materials and NP at various VC levels	EUBP, UBU	Consultancy services via the SEP agreement
Knowledge of end user markets including nano-enabled bio-based materials	UNIPD, EXE	Provision of innovation management services to clients of Ecosystem via the SEP agreement

Based on the Joint Agreement of the core partners with the SEP, significant revenue streams will exist due to the provision of services, as already analysed in the business plan. SMEs such as RDC, IRIS, ABIS, STAM, IDE, AXIA, and EXE, being members of the Ecosystem, will also create revenues providing supporting services to the SEP and the other partners based on the Joint Agreement in the field of data management, IT solutions, LCA/LCC and Safety IPR issues and legal advocacy. In addition, **End Users** (DIAD, NOVAMONT, EVERSIA, ACC and OHM) will Exploit their results in parallel. As indicated in Table 2.9, the individual exploitation routes of the End Users (TeCs) of the BIOMAC project are also presented. The five companies will have access to the PLs of BIOMAC and the other services, and thus they will be able to upscale their materials concept up to TRL 7. Through their participation they expect to strengthen their market position and provide to their client's advanced biomaterials with enhanced properties. **Preliminary business plans and financial projections have already been developed concerning a period of 5 years after the finalization of the project**. The business plans and exploitation routes will be further updated within the Ecosystem during the project in WP5 by UNIPR.

Table 2.9: *Individual KERs and exploitation routes of BIOMAC End Users*

Partner	*Core business*	*Key Exploitable Result (KER)*	*Exploitation plan/route*	*Expected turnovers*
DIAD	Automotive	3D printed Seat leverage Automotive components for interior applications	Sales of new parts to Tier 1 and OEM customers. Expand technology to other components. Product development, sales, and licensing of technology.	0.5 M € 3 jobs
NVMT	Biodegradable and compostable biomaterials	Nanoreinforced bioplastic materials biodegradable in soil for agricultural applications	Product development and sales of innovative advanced nanoreinforced bio-based materials biodegradable in soil for agricultural applications to converters. (mulch films, seeds coating, coated fertilizers granules for controlled release, etc.)	2 M € 10 new jobs
EVER	Food Packaging films	Nanopatterned PLA nanoreinforced composite Flexible food packaging	Product sales providing bio-based nanoreinforced improvement options to food packaging consumers	5 M € 25 jobs
ACC	Construction, Civil Engineering	Modular composite pedestrian bridge	Sell around 1-2 units in the first year after the completion of the project. ACCIONA also estimates that this figure could grow to 10-12 units per year after 5 years of the project completion. This means a market share of 10% in its bridge manufacturing segment.	4 M € 20 jobs
OHM	Printed Electronics	Bio based printed Electronics for wearable socks	Direct sales of integrated Printed Electrics to textile industries	1 M € 5 jobs
			Totals	**12.5 M (63 jobs)**

Based on the above analysis the end users will generate a **total revenue of 12.500.000€**. Similar to the case of the members of the Ecosystem, by assuming a new job for every 200.000€ of generated revenue, **a total of 63 new employees will be hired by the end users**. When it comes to the manufacturing industry, every new job in the sector creates 2.5 additional jobs across the supply chain. Thus, it is expected also that **160 indirect jobs** will be created across Europe.

2.2.2 Communication activities

A comprehensive communication plan is developed with the aim of providing targeted information to multiple audiences, so that awareness is created around the BIOMAC ecosystem, results and impacts at business, scientific, industrial and societal level. Whenever relevant, two-way exchange engagement will be used so to collect valuable feedback from different target audiences. Since communication addresses also audiences that are outside the BIOMAC community, activities will be differentiated in terms of message to be conveyed, medium, means and frequency of interaction. A broader, general public will be reached by top level messages that reflect the project's achievements, and particular segments of society where isolated actions would not be as effective as possible, engaging and informing the society. The campaign will use multiple channels to deliver a clear and consistent message to a variety of audiences in a way that they can be understood by non-specialists and it will have to actively engage with citizens, seeking their views and promoting a permanent dialogue. The campaign will be evaluated by using evaluation and measurement techniques including repeat of any pre-campaign research to review opinions and attitudes, as well as media analysis to examine the coverage received. Like for the dissemination measures, all the consortium partners will be involved in communication activities, whilst expertise of IBB, EUBP, RDC and EUBIA

will ensure that the quality of the material used to support these activities are of a professional level. The current dissemination and communication strategy will be updated every six months. In Table 2.3 a concrete Dissemination and Communication Strategy is summarized, and in Table 2.10 the diversity between the two activities is split including scope and means of verification. First, in order to increase the communication efficacy, supporting material will be developed to raise awareness about concepts, results, expected impacts that the BIOMAC project is expected to generate other than keeping the public informed about the BIOMAC news. This material mainly allows to reach the general audience, but it is also useful to ease the relationships with audiences that require a more focused engagement plan. Following a stakeholder analysis, the target audience of our communication activities has been defined as policy makers, training institutes, mass media (TV, radio and the written press) and multipliers for addressing the general public, as well as others. These requirements are in line with the legal basis for the Communication Strategies described in the European regulation (EC) 1828 /2006 (art.2-10). [35]

Table 2.10: *Communication plan*

Dissemination activities	***Communication activities***
Dialogues, face-to-face conversation, Group discussions, Brokerage events, Business to Clients (B2C), Round tables, Exhibitions, trade fairs, Open days, Demonstrations and prototypes, E-mail information	Newspapers and magazines, Press releases, newsletters, Manuals, Brochures, booklets, flyers Letters, Radio, Television, Video, Posters, Stickers, Banners, Billboards, Website, Policy briefs
Smaller audience, lower costs, more effort. Interactive, good for acquiring input Flexible	Potentially large audience, Uses the credibility of the mass media

Scope	***Communication Means of verification***
Identify key objectives	-- Designing the project's *graphical identity*, to ensure consistency throughout the project's duration and trademark Ecosystem to become a European reference point. -- Producing project documentation, which will stress the actual benefits of the solutions, along with incentives. -- Organising project events, to raise awareness among a targeted public, analyse project activities, establish *personalised interactions* and provide project partners with useful inputs.
Key message	The key message will be derived by the Innovation activities that will take place during the project and will be co-decided by the consortium during the project implementation. It may also be followed by supplemental and supporting messages that will enhance the central point, highlighting the project's advantages.
Campaign's communications channels	In order to reach a large audience, the consortium will try to have strong online presence. For example, the utilization of social media for the dissemination of information, and thus create Facebook, LinkedIn, Youtube and Twitter accounts from where public important information will include informative and demonstration videos.
Releasing project publications	Target is to raise awareness among relevant target audience and highlight project achievements and excellence. Resources will also be taken into consideration for the campaign's implementation for services as advertising, photography, printing, design, copywriting, editing, web and media relations management.
Releasing results to wider public	Communicating short articles in national & international newspapers, magazines, TV and on Internet platforms directed towards a general audience in mid-term, and at the end of the project.
Evaluation and redesign	During each project's progress meeting, there will be a dedicated section, where partners will attempt to review the existing communication strategy, evaluate the succeeded results and identify threats or opportunities in the area of nanotechnology.

[35] Commission Regulation (EC) 1828/2006 of 8 December 2006